Supplemental Material to "Finite spectral windows with an opening tower in seven-dimensional parents"

Draft, 3 November 2026

This is the supplement of the paper *Finite spectral windows with an opening tower in seven-dimensional parents*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

Parts contained. Lemma parts: 24, 28, 29, 30. Other parts: R11, R15, R16, R17, R18 (the letters H, L and R mark the history of the field, the literature check and the referee reports).

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

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Part 24

Commutative cylinders, spectral walls, and the two-stage compression

By Astra 6 (gpt-6-astra, effort xhigh), 3 Nov 2026; prompt in prompts/24_commutative_parent.md. Reproduced as delivered, with links made relative. The section "Numerical check" at the end is by Claude Opus 5.5. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): no error in the displayed equations, and one gap in the general statement on compression maps. The record above the section "Referee amendments" stays as delivered, and those amendments apply to it.

A commutative cylinder admits an exact realization of the **operator window** of lemmas 22 and 23 when the two finite gradings are prescribed through nonlocal spectral boundary conditions. The higher K-modes then decouple as the radius of K tends to zero. Two qualifications remain essential: these boundary conditions restrict scalar functions at the walls, and compression of scalar functions through both stages differs from generating a matrix algebra between the stages. In particular, aligned walls give $\mathbb{R} \oplus \mathbb{R}$ on the quaternionic factor under direct scalar compression, whereas the intervening matrix-algebra construction gives $\mathbb{H} \oplus \mathbb{H}$. Local boundary gradings preserve all scalar functions, but their reduction to the unchanged four-section window has a precise restriction proved below. The radial no-go of lemma 22 survives rechecking in its stated scope. For the massless one-loop potential of lemma 23, an analytic argument strengthens the numerical statement to all $n_f > 0$.

1. Theorem — An exact spectral boundary construction.

Hypotheses and conventions. Let

$$K = \mathbb{CP}^3, \qquad \mathcal{H}_K = L^2(K, \Lambda^{0,*}T^*K \otimes \mathcal{O}(1)),$$

and let

$$A = \sqrt{2}(\bar{\partial} + \bar{\partial}^*), \qquad \gamma_K = (-1)^{\deg}.$$

Use the Fubini–Study metric at a fixed reference scale. Write

$$V = \ker A \simeq \mathbb{C}^4, \qquad \Pi = P_K, \qquad \Delta_K = \inf(\operatorname{spec}|A| \setminus \{0\}) > 0.$$

Here $\gamma_K|_V = 1$. The radius parameter $r > 0$ replaces A by A/r, after the usual unitary identification of the L^2-spaces.

These are the analytic data of Example 16.9. Its S^4 option uses functions pulled back to this same twistor space. An actual operator on S^4 with the required kernel and compression map would need to be specified separately.

Let

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

On $L^2(I_\ell; \mathbb{C}^2 \otimes \mathcal{H}_K)$, set

$$\mathcal{D}_r = E\partial_x + Z\,A/r, \qquad \hat{\gamma} = Z\gamma_K.$$

All unspecified tensor factors are identities.

Choose self-adjoint involutions Γ_0, Γ_1 on V, each with two-dimensional positive and negative eigenspaces, and assume

$$-1 \notin \operatorname{spec}(\Gamma_1\Gamma_0).$$

Extend Γ_i by zero on $V^\perp$, writing $C_i = \Pi\Gamma_i\Pi$. Define $\operatorname{sgn} A = 0$ on V, and impose

$$\chi_0\psi(0) = \psi(0), \qquad \chi_\ell\psi(\ell) = \psi(\ell),$$

where

$$\chi_0 = ZC_0 - S\operatorname{sgn} A, \qquad \chi_\ell = ZC_1 + S\operatorname{sgn} A. \tag{1}$$

Then:

1. The H^1 realization of $\mathcal{D}_r$ is self-adjoint with compact resolvent and anticommutes with $\hat{\gamma}$.
2. $\Pi \otimes 1$ reduces the operator, including its domain.
3. On this reducing subspace, the operator is exactly $D(\Gamma_0, \Gamma_1)$ of lemma 23.
4. Every eigenvalue on the complementary subspace satisfies

$$|\lambda| \geq \Delta_K/r. \tag{2}$$

Proof. Put $B = SA/r$. Since $ES = Z$,

$$\mathcal{D}_r = E(\partial_x + B).$$

The operators χ_i are self-adjoint involutions and anticommute with E. Their positive eigenspaces are therefore maximal isotropic for the boundary form

$$[\langle\psi, E\phi\rangle]_0^\ell.$$

On $V^\perp$, condition (1) selects the negative spectral subspace of B at 0 and its positive spectral subspace at ℓ. These are APS conditions with the inward-normal sign reversed between the two walls. On $\ker B = \mathbb{C}^2 \otimes V$, condition (1) supplies a Lagrangian subspace:

$$\Gamma_i\psi_L = \psi_L, \qquad \Gamma_i\psi_R = -\psi_R.$$

For q in the complementary domain, integration by parts gives

$$\|\mathcal{D}_r q\|^2 = \|q'\|^2 + \|Aq/r\|^2 + [\langle q, Bq\rangle]_0^\ell \geq \|q'\|^2 + \frac{\Delta_K^2}{r^2}\|q\|^2. \tag{3}$$

The boundary term is nonnegative by the spectral choices.

An eigenbasis of A decomposes the complementary problem into finite-dimensional interval problems with maximal isotropic endpoint conditions. Each is self-adjoint. Estimate (3) identifies the domain of their self-adjoint direct sum with the stated H^1 domain and gives compactness of its resolvent. The finite-dimensional tangential kernel contributes an ordinary self-adjoint interval problem. This proves self-adjointness, compactness, and (2).

Every boundary projector commutes with Π, proving reduction. Finally, $\{\gamma_K, A\} = 0$ implies $\{\hat{\gamma}, \mathcal{D}_r\} = 0$, and $\hat{\gamma}$ preserves both boundary subspaces. ■

Define

$$K_F = \frac{1}{2\ell}\log(\Gamma_1\Gamma_0), \qquad D_F = \Gamma_0 K_F, \qquad R(x) = e^{xK_F},$$

with the principal logarithm. Let $P_\pm = (1 \pm \Gamma_0)/2$. The finite-window embedding is

$$W_\ell v(x) = \ell^{-1/2}\begin{pmatrix} R(x)P_+ v \\ R(x)P_- v \end{pmatrix}. \tag{4}$$

Direct differentiation gives

$$\mathcal{D}_r W_\ell = W_\ell D_F, \qquad \hat{\gamma}W_\ell = W_\ell\Gamma_0.$$

Thus the prescribed finite operator and grading are exact, provided $\Delta_K/r > \|D_F\|$. The next eigenvalue in absolute value is at least

$$\min\left\{\frac{\Delta_K}{r}, \frac{\pi - \ell\|D_F\|}{\ell}\right\}. \tag{5}$$

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2. Proposition — Local walls and spectral walls have different scalar domain algebras.

Hypotheses. Use the product differential expression of Result 1.

For the local option, let $G_i(y)$ be smooth self-adjoint bundle involutions and impose

$$G_i\psi_L = \psi_L, \qquad G_i\psi_R = -\psi_R \quad \text{at } x = i.$$

Assume

$$[G_i, \gamma_K] = 0, \qquad [G_i, \sigma_A(y,\xi)] = 0 \quad (\xi \in T_y^*K), \tag{6}$$

where σ_A is the tangential principal symbol.

Then these local conditions give a self-adjoint elliptic realization with compact resolvent, and **every** smooth scalar function on $K \times I_\ell$ preserves its domain.

For the spectral option (1), the maximal smooth scalar $*$-algebra preserving the domain is

$$\mathcal{A}_{\mathrm{APS}} = \{f \in C^\infty(K \times I_\ell) : f(\,\cdot\,,0), f(\,\cdot\,,\ell) \text{ are constant on } K\}\,. \tag{7}$$

The same conclusion holds for the subalgebra pulled back from $S^4 \times I_\ell$.

Proof. For the local option, the boundary involution is ZG_i. It anticommutes with the normal matrix E. Condition (6) makes it anticommute with the principal symbol of the adapted tangential operator SA/r. Hence its positive eigenspace is an elliptic Lagrangian boundary condition. The corresponding local ellipticity and adjoint-domain criteria are described in Bär–Ballmann, Corollary 4.18 and Example 4.20. Primary reference

Scalar multiplication commutes pointwise with G_i, so it preserves the endpoint conditions and the H^1 domain. Moreover,

$$[\mathcal{D}_r, f] = E\,\partial_x f + Z[A, f]/r$$

is bounded for each $r > 0$.

For spectral walls, preservation by a scalar $*$-algebra is equivalent to

$$[M_{f_i}, \chi_i] = 0.$$

Independence of the spin matrices Z, S gives

$$[M_{f_i}, C_i] = 0, \qquad [M_{f_i}, \operatorname{sgn} A] = 0.$$

Since $C_i^2 = \Pi$, the first equation implies

$$[M_{f_i}, \Pi] = 0. \tag{8}$$

Consequently multiplication by f_i preserves $H^0(K, \mathcal{O}(1))$. If s is such a section, then

$$0 = \bar\partial(f_i s) = (\bar\partial f_i)s.$$

The sections of $\mathcal{O}(1)$ have no common zero, so $\bar\partial f_i = 0$ everywhere. A holomorphic function on compact connected $\mathbb{CP}^3$ is constant. Constants plainly preserve (1), proving (7). ■

The hypotheses in (6) are a sufficient local ellipticity condition. An arbitrary bundle grading supplies a pointwise isotropic condition, but self-adjointness on an H^1 domain and compact resolvent still require an ellipticity check. Likewise, prescribing a grading only on V leaves the boundary problem incomplete until a condition on $V^\perp$ is supplied.

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3. Proposition — Compression of a local grading and the unchanged level-one window.

Hypotheses. Let G be a self-adjoint bundle involution acting by multiplication, and let

$$g = \Pi G \Pi|_V, \qquad Q_K(f) = \Pi M_f \Pi|_V.$$

Keep the level-one space and scalar representation of Example 16.9 unchanged.

Then

$$1_V - g^2 = ((1-\Pi)G\Pi)^*((1-\Pi)G\Pi). \tag{9}$$

In particular, g is a grading exactly when $[G, \Pi] = 0$. In that event,

$$[g, Q_K(f)] = 0 \quad \text{for every scalar } f. \tag{10}$$

For either the full $\mathbb{CP}^3$ scalar algebra or the S^4 pullback algebra, this forces

$$g = 1_V \quad \text{or} \quad g = -1_V. \tag{11}$$

Proof. Expand $\Pi G^2 \Pi = \Pi$ using $1 = \Pi + (1 - \Pi)$; this gives (9). Vanishing of its positive right-hand side implies $(1 - \Pi)G\Pi = 0$, and self-adjointness gives the other off-diagonal block.

Since scalar multiplication commutes with G, reduction then gives (10). The compressed $\mathbb{CP}^3$ functions generate $M_4(\mathbb{C})$. The compressed real S^4 functions generate $M_2(\mathbb{H})$, whose complex-linear commutant on $\mathbb{C}^4$ is also $\mathbb{C}1$. Thus a self-adjoint involution satisfying (10) is ± 1. ■

For **aligned local walls** with $[G, A] = 0$, the finite zero-mode window is explicitly

$$v \longmapsto \ell^{-1/2} \begin{pmatrix} P_+(g)v \\ P_-(g)v \end{pmatrix}.$$

The induced grading is g, and scalar compression is

$$P_+(g)Q_K(f)P_+(g) + P_-(g)Q_K(f)P_-(g) = Q_K(f).$$

By (11), the unchanged four-section window has trivial grading. Its generated algebra remains $M_2(\mathbb{H})$ or $M_4(\mathbb{C})$, according to the scalar algebra chosen. The globally constant choices $G = \pm 1$ give concrete self-adjoint examples with this window and a separated higher spectrum.

There is also an explicit local field illustrating failure of reduction. Add a trivial auxiliary $\mathbb{C}^4$ bundle and put

$$G(y) = \sum_{a=1}^{5} y_a \Gamma_a^{\text{aux}},$$

where $y \in S^4$ is the twistor projection and the Γ_a satisfy $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$. This is a local involution satisfying (6), but now the tangential kernel has dimension 16, and

$$g = \frac{1}{5} \sum_{a=1}^{5} \Gamma_a^{\text{section}} \otimes \Gamma_a^{\text{aux}}.$$

Using $\operatorname{tr}(\Gamma_a \Gamma_b) = 4\delta_{ab}$,

$$\operatorname{tr}(g^2) = \frac{16}{5} \neq 16.$$

Thus g is not a grading. Equation (9) even gives

$$\|(1 - \Pi)G\Pi\| \geq \frac{2}{\sqrt{5}},$$

independently of r. This is a bound on boundary-trace mixing; it does not identify the actual low spectral window of the coupled local problem.

A local $G(y)$ specifies a **field of gradings**. It does not select a marked point p. In the tautological quaternionic line bundle,

$$\Pi_y = \frac{1 + y_a \Gamma_a}{2},$$

the natural field $2\Pi_y - 1$ acts as $+1$ on the fibre $\operatorname{ran} \Pi_y$. By contrast, the finite grading

$$\Gamma_p = p_a \Gamma_a = 5Q_K(p \cdot y)$$

uses a separately chosen point p. Its action on global section coefficients is nonlocal.

The restriction proved here concerns the unchanged level-one space, a local multiplication grading, and its compression as an actual involution. A different coupled window, additional fibres, or spectral boundary conditions remain available directions.

4. Theorem — The composed compression map and the aligned-wall algebra.

Hypotheses. Use the exact spectral construction of Result 1. For a scalar multiplier $f(y,x)$, let

$$T_f(x) = Q_K(f(\,\cdot\,,x)), \qquad \mathcal{E}_0(T) = \frac{1}{2}(T + \Gamma_0 T \Gamma_0).$$

Use real-valued functions and real algebra generation when discussing $M_2(\mathbb{H})$.

Then the compression directly from the scalar cylinder to the finite window is

$$\boxed{\Phi_\ell(f) = \frac{1}{\ell}\int_0^\ell \mathcal{E}_0\big(R(x)^* T_f(x) R(x)\big)\, dx.} \tag{12}$$

It is independent of r.

For aligned walls $\Gamma_1 = \Gamma_0 = \Gamma_p$, direct scalar compression on the S^4 factor generates

$$\mathbb{R} \oplus \mathbb{R}. \tag{13}$$

If instead one first generates $M_2(\mathbb{H})$ and then applies the interval compression, the result is

$$\mathbb{H} \oplus \mathbb{H}. \tag{14}$$

Proof. Insert (4) on both sides of the multiplication operator. Orthogonality of the two interval spin components gives exactly (12).

For real S^4 functions, Example 16.9 implies

$$Q_K(C^\infty(S^4, \mathbb{R})) = \mathrm{span}_{\mathbb{R}}\{1, \Gamma_1, \dots, \Gamma_5\}. \tag{15}$$

Indeed all compressed functions are quaternionic Hermitian matrices; that space has real dimension six, and the displayed six matrices belong to the image and are independent.

For aligned walls $R = 1$. Clifford multiplication gives

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p.$$

Therefore the image of direct scalar compression is

$$\mathrm{span}_{\mathbb{R}}\{1, \Gamma_p\}.$$

This space is already the algebra $\mathbb{R} \oplus \mathbb{R}$, proving (13).

On the other hand,

$$\mathcal{E}_p(M_2(\mathbb{H})) = \{a \in M_2(\mathbb{H}) : [a, \Gamma_p] = 0\} = \mathbb{H} \oplus \mathbb{H},$$

proving (14). The noncommuting products needed to generate the quaternionic diagonal blocks have been inserted between the two compressions in this second procedure. ■

On the single $\mathbb{CP}^3$ factor, real scalar compression is onto the Hermitian 4×4 matrices. A chosen $2+2$ aligned grading therefore gives all Hermitian block-diagonal matrices and generates

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

For the **chosen two-sheet coarse correspondence** of Example 16.9, with

$$H_0 = \mathrm{Hom}(V, W) \oplus \mathrm{Hom}(W, V), \qquad \Gamma_0(\xi, \eta) = (\Gamma_p \xi, -\eta \Gamma_p),$$

the same calculation gives

$$\boxed{\mathrm{alg}_{\mathbb{R}}(\text{direct aligned scalar compression}) = \mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).} \tag{16}$$

The V-side matrices act on the left of η and commute with its right grading, so that factor survives unchanged.

Endpoint restrictions do not change these finite-dimensional images. If a linear functional vanishes on compressions of functions supported in $0 < x < \ell$, its smooth matrix-entry density vanishes in the interior and hence everywhere. Finite-dimensional separation then gives equality with the unrestricted image, exactly as in Result 7 of lemma 20.

At the intermediate stage, however, spectral walls still give scalar endpoint values. Even after algebra generation, those values remain scalar. Enlarging them to the full stabilizers of Γ_0, Γ_1, as in lemma 23, is another change of the parent algebra. Equality of finite compression images does not identify the parent gauge groups or one-forms.

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5. Theorem — Radius collapse, interval collapse, and controlled mixing.

Hypotheses. First use the exact construction of Result 1. Identify the varying Hilbert spaces by metric rescaling. For resolvent formulas, fix units in which the nonreal spectral parameter is i.

Then, at fixed ℓ,

$$\left\| (\mathcal{D}_r - i)^{-1} - \iota_\Pi (D(\Gamma_0, \Gamma_1) - i)^{-1} \iota_\Pi^* \right\| \le \frac{r}{\Delta_K}. \tag{17}$$

Thus radius collapse gives the entire matrix interval operator in generalized norm-resolvent sense.

If D_F is fixed, $\ell \to 0$, and

$$\Gamma_1(\ell) = e^{\ell \Gamma_0 D_F} \Gamma_0 e^{-\ell \Gamma_0 D_F},$$

then $r \to 0$ and $\ell \to 0$ decouple both towers and leave the finite operator D_F. No relation between the two rates is required in this exactly reducing construction.

Proof. The resolvent is block diagonal with respect to Π. Estimate (2) bounds its complementary block by r/Δ_K, proving (17). For the finite-window limit, (5) bounds the complementary resolvent by

$$\max\left\{ \frac{r}{\Delta_K}, \frac{\ell}{\pi - \ell \|D_F\|} \right\},$$

which tends to zero. ■

A useful mixing statement requires additional hypotheses. Suppose a transformed problem has a common domain and the form

$$\widetilde{\mathcal{D}}_r = \mathcal{D}_r + V_r, \qquad V_r = V_r^*, \qquad \|V_r\| \le b_r.$$

Write

$$v_r = \|(1 - \Pi) V_r \Pi\|.$$

For an eigenvector $\psi = p + q$, with $p = \Pi\psi$, $q = (1 - \Pi)\psi$, and $|\lambda| \le L$, assume

$$g_r = \Delta_K / r - b_r - L > 0.$$

Then

$$\boxed{\|q\| \le \frac{v_r}{g_r} \|p\|, \qquad \|\text{Schur correction at } \lambda\| \le \frac{v_r^2}{g_r}.} \tag{18}$$

Proof. The complementary eigenvalue equation gives

$$q = -(\mathcal{D}_{r,Q} + V_{QQ} - \lambda)^{-1} V_{QP} p.$$

The inverse has norm at most g_r^{-1}. Substituting this formula into the p-equation gives the second bound. ■

For bounded b_r, v_r, both errors are $O(r/\Delta_K)$. If a common-domain construction instead produces terms of order $1/\ell$, sufficient conditions include

$$\frac{rv_r}{\Delta_K} \to 0, \qquad \frac{rv_r^2}{\Delta_K} \to 0,$$

together with convergence of the Π-diagonal perturbation.

An arbitrary local $G_i(y)$ generally fails $[G_i, \Pi] = 0$. Although the differential expression commutes with Π, its domain then does not. Such a boundary change is not automatically a bounded perturbation satisfying (18). The explicit local field in Result 3 has nonvanishing boundary mixing at every radius. No convergence to the prescribed two-gradings operator is established here for that nonreducing local problem.

At the level of observables, radius collapse leaves (12) unchanged. For an x-independent $T = Q_K(f)$,

$$\|\Phi_\ell(f) - \mathcal{E}_0(T)\| \le \ell \|D_F\| \, \|T\|. \tag{19}$$

If T is even for Γ_0, the stronger formula of Theorem 22.4 applies:

$$\Phi_\ell(f) = \operatorname{sinc}(\ell \operatorname{ad}_{D_F})(T),$$

$$\|\Phi_\ell(f) - T\| \le \frac{\ell^2}{6} \|[D_F, [D_F, T]]\|. \tag{20}$$

Equation (19) follows by integrating $\|R^* T R - T\| \le 2x \|D_F\| \|T\|$; equation (20) follows by taking the even part of the conjugation series.

Consequently there is convergence of the **compression maps on their scalar operator systems**. The limit is $\mathcal{E}_0 Q_K$. It does not turn that image into every element of $\mathcal{E}_0(P)$; Result 4 computes the difference explicitly. Algebra generation remains a separate operation.

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6. Proposition — The second sheet, reality, even algebra, and connector remain prescribed data.

Hypotheses. Consider the scalar cylinder construction above and, separately, its extension using the two-sheet coarse correspondence of Example 16.9.

The resulting distinctions are:

Structure	What follows under the stated choices
Second sheet and the two Hom spaces	They are inputs in the coarse correspondence. A single four-section cylinder window does not supply them.
$J_1^2 = -1$, $J_1 \Gamma_i = -\Gamma_i J_1$	These are additional compatibility conditions on the prescribed gradings and antiunitary.
$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$	This is the derived even part after choosing the quaternionic real form, the grading, and intermediate algebra generation. Direct aligned scalar compression gives (16).
Connector and $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	The connector's lines and coupling are chosen. Their intertwining algebra is then derived.

Proof. The one-factor construction starts with $V = \mathbb{C}^4$, and (4) has domain V. The two-sheet construction instead starts with

$$(W \otimes \overline{V}) \oplus (V \otimes \overline{W}).$$

Geometrically, its analytic realization uses product factors and a restricted scalar algebra, as Theorem 16.7 specifies. It is additional correspondence data over the scalar spaces.

For reality, suppose an antiunitary $\mathcal{J}_1$ on the tangential Hilbert space satisfies

$$\mathcal{J}_1^2 = -1, \quad \mathcal{J}_1 A = -A\mathcal{J}_1, \quad [\mathcal{J}_1, \gamma_K] = 0,$$

and its restriction J_1 to V anticommutes with both Γ_i. Then

$$\widehat{J} = E\mathcal{J}_1$$

preserves (1) and obeys

$$\widehat{J}^2 = 1, \qquad [\widehat{J}, \mathcal{D}_r] = 0, \qquad \widehat{J}\hat{\gamma} = -\hat{\gamma}\widehat{J}.$$

On the window,

$$J_F = \Gamma_0 J_1.$$

These identities follow directly from the stated commutation relations. They specify sufficient conditions; the natural $\mathcal{O}(1)$ cylinder has not supplied such an extension or its order-zero property here.

In particular, the quaternionic antiunitary j used to recognize $M_2(\mathbb{H})$ commutes with the Γ_a. It does not itself satisfy the required anticommutation with Γ_p.

The even-algebra assertion is the centralizer calculation in Result 4 after intermediate generation.

Finally, choose

$$w \in W_+, \quad v \in V, \quad \|w\| = \|v\| = 1, \quad \kappa \neq 0,$$

and prescribe

$$T_\kappa \xi = \kappa(w^* \xi v)\, v w^*.$$

Its intertwining conditions require the lines $\mathbb{C}w, \mathbb{C}v$ to be reducing, with the same scalar action. A quaternion preserving $\mathbb{C}w$ lies in its chosen copy of $\mathbb{C}$; $v^\perp$ carries arbitrary $M_3(\mathbb{C})$, and W_- carries arbitrary $\mathbb{H}$. Hence

$$\mathcal{A}(T_\kappa) = \{(\lambda \oplus q, \lambda \oplus m)\} \simeq \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}),$$

as in Result 2 of lemma 20. ■

A marked $p \in S^4$ fixes a quaternionic line, but choosing w also requires a complex line within it. The second line v, the coupling, and an interface realizing the connector remain necessary. Scalar agreement at marked points has unchanged compression image by Result 7 of lemma 20, so it supplies none of this selection by itself.

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7. No-go theorem — Recheck of the radial statement in lemma 22.

Hypotheses. Let the parent be exactly the one-interval model of lemma 22: fixed wall coupling X, zero bulk mass, the even algebra $\mathcal{A}_\ell$, scalar inner fluctuations Eh with $h^* = -h$, and the **moving spectral window**. For real fluctuations, include the conjugate one-form and assume the stated order-zero representation.

Then every fluctuated finite operator has the same singular values as D_F. This is an exact finite-amplitude spectral statement; it uses no action or loop expansion.

Proof. Solve

$$u' = -hu, \qquad u(0) = 1, \qquad w = u(\ell),$$

and set $v(x) = u(x)w^*$. Then

$$v(\ell) = 1, \qquad v(0) = w^*.$$

Since h is even, w preserves the boundary splitting at 0; the identity at ℓ preserves the fixed coupled wall. Thus v preserves the original domain, and

$$v D_X v^* = D_X + Eh.$$

For the real fluctuation, the analogous implementing unitary is

$$vJ_F vJ_F^{-1};$$

order zero separates the two factors in its derivative.

The full operators are unitarily equivalent, so their moving spectral windows are unitarily equivalent. In the finite identifications of lemma 22, the operators are $w^* D_F w$, or $U_w^* D_F U_w$ in the real case. Their singular values coincide. ■

The moving-window hypothesis matters. For example, with $D_F = m\sigma_1$ and $h = i\sigma_3$, fixed compression gives

$$Q_\ell(Eh) = -\frac{1-\cos(2m\ell)}{2m\ell}\sigma_2.$$

Adding a multiple of this operator to the fixed $m\sigma_1$ changes its singular values at second order.

The verdict agrees with the author's amended theorem and Terra's independent endpoint-preserving proof. In the usual nondegenerate Standard Model representation, the normalized shifted Higgs quaternion lies on the unit-quaternion orbit; its norm is fixed.

The known escapes are a variable wall coupling, odd connection directions, suitable bulk masses, additional internal directions with nontrivial connection data, or additional elementary or composite fields. A quantum effective action on the same configuration space does not add a radial coordinate. The proposed $K \times I_\ell$ parent has tangential directions and tangential one-forms, so this theorem does not decide its full fluctuation space.

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8. Lemma — A logarithmic concavity property of the one-loop integral.

Hypotheses. Let

$$\mathcal{J}(z) = \int_0^\infty \frac{x^3}{\sinh^2 x + z}\, dx, \qquad z > 0,$$

and define

$$e(z) = -\frac{z\mathcal{J}'(z)}{\mathcal{J}(z)}.$$

Then $0 < e(z) < 1$, and e is nondecreasing.

Proof. Positivity and

$$0 < \frac{z}{\sinh^2 x + z} < 1$$

show that $e(z)$, expressed as a weighted average of this fraction, lies strictly between zero and one.

Put $x = e^v$, $z = e^u$. Then

$$\mathcal{J}(e^u) = \int_{\mathbb{R}} \exp\left(4v - \log(\sinh^2(e^v) + e^u)\right)\, dv. \tag{21}$$

The function

$$b(v) = 2\log\sinh(e^v)$$

is convex, because, with $x = e^v$,

$$b''(v) = 2x(\coth x - x\ \operatorname{csch}^2 x) > 0.$$

The last inequality is equivalent to $\sinh x \cosh x > x$.

The function $(u, v) \mapsto \log(e^{b(v)} + e^u)$ is therefore convex. The integrand in (21) is jointly log-concave. Marginal integration preserves log-concavity by Prékopa's theorem, so $u \mapsto \log \mathcal{J}(e^u)$ is concave. We use precisely Theorem 6, p. 342, of Prékopa's original paper.

Finally,

$$\frac{d}{du}\log \mathcal{J}(e^u) = -e(e^u).$$

Concavity makes this derivative nonincreasing, proving the assertion. ■

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9. Theorem — The massless one-loop potential has no interior local minimum.

Hypotheses. Let the action be exactly the one-loop determinant displayed in lemma 23: a flat interval, zero bulk masses, frequency-one fermions, the stated charged and neutral gauge towers and multiplicities, and the prescribed connector wall condition. Assume that no fermion connector modifies the displayed fermion tower, and add no wall potential or other angle-dependent term.

Let $n_f > 0$, set $t = \sin^2\theta$, and write

$$\mathcal{V}(t) = 8\pi^2\ell^4 V(\theta).$$

Then:

1. $t = 1$, equivalently $\theta = \pi/2$, is the unique global minimum.
2. If $n_f \geq 54/5$, the potential is strictly decreasing on the interior.
3. If $0 < n_f < 54/5$, there is exactly one interior stationary point, and it is a strict maximum.

Proof. Define

$$F(z) = \int_0^\infty x^3 \log\left(1 + \frac{z}{\sinh^2 x}\right) dx, \qquad F'(z) = \mathcal{J}(z).$$

The exact potential is

$$\mathcal{V}(t) = 3F(2t(1-t)) + \frac{3}{2}F\Big(\frac{16}{5}t(1-t)\Big) - n_f F(t). \tag{22}$$

Its gauge terms are nonnegative and vanish at $t = 1$. Since F is strictly increasing, the fermion term is uniquely minimized there. This proves item 1.

Differentiate:

$$\mathcal{V}'(t) = (1-2t)\left[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\Big(\frac{16}{5}t(1-t)\Big)\right] - n_f\mathcal{J}(t). \tag{23}$$

For $t \geq 1/2$, this is strictly negative.

For $0 < t < 1/2$ and $c \geq 2$, put

$$R_c(t) = (1-2t)\frac{\mathcal{J}(ct(1-t))}{\mathcal{J}(t)}.$$

Let $a = ct(1-t)$. Then $a \geq t$, and Result 8 gives $e(a) \geq e(t)$. Consequently,

$$\begin{aligned}\frac{d}{dt}\log R_c(t) &= -\frac{2}{1-2t} - \frac{1-2t}{t(1-t)}e(a) + \frac{e(t)}{t} \\ &\leq -\frac{2}{1-2t} + \frac{e(t)}{1-t} \\ &< -\frac{2}{1-2t} + \frac{1}{1-t} < 0.\end{aligned}$$

Thus every R_c is strictly decreasing. Since $\mathcal{J}(0)$ is finite and positive,

$$R_c(0+) = 1, \qquad R_c(1/2-) = 0.$$

Therefore

$$H(t) = 6R_2(t) + \frac{24}{5}R_{16/5}(t)$$

decreases strictly from 54/5 to zero. Equation (23) becomes

$$\mathcal{V}'(t) = \mathcal{J}(t)\,[H(t) - n_f] \quad (0 < t < 1/2).$$

For $n_f \geq 54/5$, its sign is negative. For $0 < n_f < 54/5$, it changes sign exactly once, from positive to negative. Since $t = \sin^2\theta$ is strictly increasing on the interior, the same stationary-point classification holds in θ. ■

This proves the massless statement for the entire parameter range, including the previously numerical range. No numerical computation is used in this proof.

The recorded author and Terra referee calculations agree on the sampled massless cases. The logarithmic-concavity proof above is an additional single-agent derivation; it has not received a fresh independent model review in this turn. The recorded agreement therefore should not be described as a multi-model review of this new proof.

The escapes remain changes to the stated determinant: frequency-two fermions, modified form factors through warping or other geometry, suitable bulk-mass sectors, wall potentials, and wall kinetic or interaction terms. The full effective action of the proposed $K \times I_\ell$ parent, including its K-tower, has not been computed here.

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10. Proposition — Scope of the bulk-mass scans.

Hypotheses. Retain the flat gauge terms of (22) as an independent assumption. Replace the fermion factor by the expression in lemma 23:

$$1 + t(G(x) - 1), \qquad G(x) = \frac{(w_Q \coth w_Q - c_Q)(w_U \coth w_U - c_U)}{x^2}, \quad w_i = \sqrt{x^2 + c_i^2}.$$

If $c_U = -c_Q = c$, then the potential is finite and $t = 1$ is its unique global minimum for every $n_f > 0$. The reported absence of additional interior **local** minima at nonzero c remains a numerical statement here.

If $a = c_Q + c_U \neq 0$, the displayed unrenormalized integral has angle-dependent ultraviolet divergences.

Proof. For opposite masses,

$$G(x) = 1 + \frac{w^2}{x^2 \sinh^2 w}, \qquad w = \sqrt{x^2 + c^2}.$$

The added fraction is positive and decays exponentially at infinity. The negative fermion logarithm is strictly decreasing in t. The gauge logarithms are nonnegative and vanish at $t = 1$, proving the global-minimum assertion.

For general masses,

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

and hence

$$\log(1 + t(G - 1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by the integration measure $x^3 dx$ produces angle-dependent divergences. A finite potential therefore requires a specified completion, regulator, and counterterms. ■

The opposite-mass scan in lemma 23 samples finitely many masses, multiplicities, and angles. It cannot exclude unsampled local minima. General bulk masses and altered gauge form factors remain directions for constructing an interior vacuum; these conclusions leave that question open.

Consequence for T0 and T0b

For T0, a commutative scalar representation with prescribed spectral walls gives an exact interval operator window and controlled tower decoupling, while local walls preserve the full scalar algebra under the ellipticity hypotheses above. The derivation of a nontrivial local grading and the desired algebra from one coupled scalar window remains open, and intermediate algebra generation must be stated explicitly. For T0b, the

second sheet, compatible reality, quaternionic grading choice, and connector remain inputs; their algebraic consequences can be derived once those choices are supplied. Neither the radial theorem nor the specified one-loop potential closes the commutative-parent problem.

Referee amendments

From Terra 6.1. These apply to the text above.

- **Result 2.** Write G_0 at x = 0 and G_l at x = l. Equation (6) supplies the principal-symbol condition of Bar-Ballmann, Corollary 4.18 and Theorem 4.11; their Example 4.20 assumes more. On the unaugmented Dolbeault bundle, the Clifford representation is irreducible, so a fibre endomorphism commuting with every sigma_A is scalar, and a smooth involution of this form is +1 or -1. Additional fibres and other local elliptic projectors lie outside this sufficient criterion.
- **Result 3.** The statement on marked points concerns the natural field displayed there. A local field can encode a chosen point: on the auxiliary bundle, G_p(y) = p_a Gamma_a^aux is a local constant field whose coefficients contain p.
- **Result 4.** Equation (16) has the two Hom spaces and the two scalar algebras of Example 16.9 as inputs. The endpoint argument proves equality of the finite-dimensional linear images, and does not identify the parent algebras, norms, gauge groups or one-forms.
- **Result 5, gap in the final assertion.** Convergence of compression maps needs a common domain and a collapse map. With aligned walls, D_F = 0 and f_l(y, x) = 6 (x/l)(1 - x/l), both endpoint values vanish while Phi_l(f_l) = 1. A correct form: for F(y, s) fixed on K x [0, 1] and f_l(y, x) = F(y, x/l), Phi_l(f_l) tends to E_0 Q_K(int_0^1 F(., s) ds). For x-independent ambient multipliers the limit is E_0 Q_K, by (19). The rate discussion after (18) also needs control of the denominator: a sufficient addition is r (b_r + L) <= (1 - epsilon) Delta_K. The case b_r = O(1/l) follows from the stronger estimate |D_{r,Q}| >= (Delta_K^2/r^2 + pi^2/(4 l^2))^{1/2}, from the one-ended Poincare inequality on the positive and negative B-spectral components.
- **Result 6.** At the finite level, if the gradings are Gamma_p, Gamma_q, a unit n orthogonal to p and q gives J_1 = Gamma_n j, with J_1^2 = -1 and the required anticommutation. Extending this choice to the parent remains a separate task.
- **Lemma 8 and Theorem 9.** An independent check of the key monotonicity, without log-concavity or Prekopa's theorem: with rho(s) = (arsinh sqrt(s))^3/(2 sqrt(s (1 + s))), J(z) = int_0^infinity rho(s)/(s + z) ds; with L(v) = log rho(e^v), L'(v) = 3 tanh(x)/(2x) - (1 + tanh^2 x)/2 for v = 2 log sinh x, which is strictly decreasing. For z = e^u, with the probability density in y proportional to e^y rho(e^{u+y})/(1 + e^y), e(e^u) is the mean of 1/(1 + e^y), and its derivative in u is Cov(1/(1 + e^y), L'(u + y)) > 0, as both are strictly decreasing. Theorem 9 is for theta in [0, pi/2]. At theta = pi/2 the determinant has its continuous value, while the principal-logarithm window is degenerate. Its hypotheses are the flat massless one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no other angle-dependent term.
- **Result 10.** The divergence is stated for nonzero fermion multiplicity, with no cancellation by additional sectors. The opposite-mass scan in lemma 23 is a numerical result on its sampled parameters.

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a temporary directory, not kept: E_p(Gamma_a) = p_a Gamma_p to 10^-15, which gives equation (13); tr(g^2) = 16/5 for the auxiliary field of Result 3; on grids, e(z) lies in (0, 1) and is nondecreasing for z in [10^-3, 50]; H(t) is strictly decreasing on (0, 1/2), with H near 54/5 at small t and near 0 at t = 1/2; for n_f in {1, 3, 6, 8, 10}, H - n_f has one sign change, with the stationary angle 0.34 at n_f = 8, which agrees with the referee of lemma 23. Results 1, 2, 5 and 6 were read and not separately computed.

Part 28

Two sheets as the two ends of one connected commutative parent

For STRATEGY T0 and T0b, following lemma 16 (Example 16.9), lemma 24 (Proposition 6) and lemma 09. Propositions 28.1 to 28.4 and 28.13 are by Claude Opus 5.5. Results 28.8 to 28.12 and 28.14 to 28.21 are by Astra 6, job 28 (report, with the full proofs; prompt), checked by Claude Opus 5.5 by hand and in Python. Proposition 28.13 and the numerics are refereed by Terra 6.1 (report), with its corrections applied. The window is a conjecture with numerical support; the numbers come from a radial solve in Python (numpy and scipy), described in the section Numerical check; the script is not kept.

Correction, 3 Nov 2026. The first version of Result 28.5 and of the numbers in Remark 28.7 (commits dd1b14c, 530fa06) used the wrong sign of the spin connection in the Python slice operator. That changed the singular values and the boundary fibre parts, so those numbers are withdrawn, including the next singular value 1.04, the value 1.35 on the representation 16, and the boundary fibre parts -0.011 to -0.853. The exponents at the origin and the multiplicities are unaffected. Astra 6 did not reproduce the old boundary values and its proofs do not use them. We found the error when that operator gave a core zero mode in the representation 20, against the exact count of Proposition 28.11. The singular values, fibre parts and densities below are recomputed, and the script tests the operator against exact statements: {A_0, A_1} = 0, A_0^2 equal to the Casimir formula on S^4, D^2 on (0, 0)-forms with values in O(1) at the Fubini-Study radius equal to j(j + 4), and A_2 equal to its curvature contraction (the last test proposed by Terra 6.1).

Update, 4 Nov 2026. Lemma 29 proves Conjecture 28.6 in case (b) for the metric class of its Setting 29.0, and in case (a) at the Fubini-Study radius, refereed by Terra 6.1. Item 3 below and the Status of Proposition 28.13 describe the position before that lemma.

Outcome.

1. *The mechanism is proved at the level of compression maps.* A smooth function on the disc bundle X has independent values on the zero section S^4 and on the boundary CP^3, and at the zero section it is a function on S^4 alone (Proposition 28.8). For prescribed windows over this X, real functions generate exactly M_2(H) + M_4(C) at every localization scale (Theorem 28.21). In Example 16.9 the second sheet and the restriction to S^4 were inputs, so this is a connected source for them.
2. *The two ends of an actual local operator are understood.* There is a self-adjoint operator with a local wall (Theorem 28.9). The core carries exactly four states in the representation 4 (Proposition 28.11). At the wall, large mass keeps a tangential mixing (Lemma 28.12). For the pull-back connection the boundary mode then has the fibre part -w(rho) in closed form, with w(rho) = 2(rho^2+2)^2/(9+2(rho^2+2)^2), which reaches the level-one value of Example 16.9 only as rho -> infinity (Proposition 28.13). This is a statement on the finite wall model, and its convergence from the full operator is numerical. A connection preserving the line summands in a collar, with the Fubini-Study metric at the wall, gives the level-one value in that model.
3. *The isolated window of the whole operator stays a conjecture* (Conjecture 28.6): what is missing is the normal reductions with their errors, the matching, and a coercive estimate uniform in the mass and in all representations of Sp(2) (Lemma 28.14).
4. *The algebra M_2(H) + M_4(C) is the algebra of the limit map.* Generated algebras need not converge when compression maps do (Counterexample 28.17). At the sample points resolved numerically, the core mode has a nonzero density breaking the quaternionic structure, empirically of order lambda^2; if it is nonzero in the continuum, Proposition 28.16 gives M_4(C) for the core block at those masses. This is numerical evidence and does not establish the exact algebra at any finite mass.

Setting

Let G = Sp(2), K = Sp(1)_a x Sp(1)_b and H = Sp(1)_a x U(1)_b, so that G/K = S^4, K/H = S^2 and G/H = CP^3. Let S be the bundle G x_K (1, 2), so that CP^3 = P(S). The normal bundle of the zero section is G x_K (1, 3), and away from it X = (0, 1] x CP^3, with radial coordinate x and S^2 fibres of radius rho(x), rho(x) ~ x at x = 0. Charges of U(1)_b are normalized so that the doublet of Sp(1)_b

has +-1. The isotropy representation is (2)_{+-1} (+) (1)_{+-2}. The spinors of X restrict on a slice to Sigma_6 = Sigma^+ (+) Sigma^-, with c(d_x) = Z the chirality of Sigma_6, where

```
Sigma^+ = (1)_2 (+) (1)_0 (+) (2)_{-1},   Sigma^- = (2)_1 (+) (1)_0 (+) (1)_{-2}.
```

The representation 4 of G restricts to (2)_0 (+) (1)_1 (+) (1)_{-1}, that is (2, 1) (+) (1, 2) of K. Let Gamma_a be the Clifford generators on 4, with Gamma_5 commuting with K and equal to +-1 on the two summands, and let x_a(g) be defined by lambda(g) Gamma_5 lambda(g)^-1 = x_a(g) Gamma_a.

Proposition 28.1 (the twist).

1. For k != 0, O(k) on CP^3 does not extend to a line bundle on X, since H^2(X) = H^2(S^4) = 0 while O(k) has degree k on the fibre S^2.
2. The pullback of S to X restricts on CP^3 to O(1) (+) O(-1), as the tautological splitting. So E = S^3(S), which is pulled back from S^4 and has trivial connection along the fibres, restricts to O(3) (+) O(1) (+) O(-1) (+) O(-3). The summand O(3) gives the Dolbeault operator on O(1) of Example 16.9, since Sigma (x) O(3) = Lambda^{0,*} (x) O(1).
3. On E the section Phi(v) = phi(|v|) (v-hat . T), with T the spin-3/2 generators and phi odd and smooth, is smooth and G-equivariant, with zero set exactly S^4. In the frame of the slice it is phi(x) tau, with tau diagonal in the summands, with eigenvalues +-3/2, +-1/2.

So we take the Callias-type operator T = D_X(E) + i Phi on spinors of X tensored with E, and the self-adjoint operator D-hat = [[0, T^*], [T, 0]], with the grading Gamma-hat = diag(1, -1). At x = l we take the local condition Gamma-hat Z = -1 on the doubled spinor, that is u_even(l) = 0 for T and v_odd(l) = 0 for T^*. It is a chirality condition, elliptic and local, and it preserves the domain under all smooth functions on X.

Proposition 28.2 (multiplicities). On a slice, Hom_H(4, Sigma^+ (x) E) and Hom_H(4, Sigma^- (x) E) each have dimension 5. For the summand O(3), the dimensions are 1 and 0. So on 4, the operator on the summand O(3) with its U(1) connection vanishes for every G-invariant metric on the slice, and its kernel contains H^0(CP^3, O(1)) = 4 for every fibre radius. This concerns the separate line connection. The pull-back connection of E does not preserve the summands, and its slice operator is invertible for small fibres (Proposition 28.10).

Proof. Tensoring the lists above with the charges of E gives the multiplicities. The operator is odd under Z and maps the multiplicity space of 4 into itself. ■

Proposition 28.3 (the zero section). The fibre operator c_N . nabla + i Phi on R^3 with values in E has zero modes (1, 3) (+) (1, 1) of K, by the Callias count for the positive weights 3/2 and 1/2, with one sign of Phi. On S^4 this gives the Dirac operator twisted by (1, 3) (+) (1, 1). Then Hom_K(4, S^- (x) (1, 3)) has dimension 1, through (1, 2), and Hom_K(4, S^+ (x) (1, 3)) = 0, so the twisted operator vanishes on that copy of 4, in agreement with its index 4. The untwisted part (1, 1) has eigenvalue 2 on the representation 4. Every map in Hom_K(4, S^- (x) (1, 3)) annihilates (2, 1) in 4. These are the twistor spinors of S^4, as sections of S.

Proof. The representation count is as in Proposition 28.2, and the Callias count is a reference, with the exact count in Proposition 28.11. For the last sentence, the first-order operator from (1, 2) to (2, 3) in the slice limit is the twistor operator, and its kernel is the 4. ■

On a representation V the twisted operator has D^2 = Cas_G(V) + 3/2 - Cas_K(tau), for the twist tau, with Cas_G(4) = 5/2, Cas_G(16) = 15/2, Cas_G(20) = 21/2, Cas_G(40) = 29/2, Cas_K(j_a, j_b) = 2 j_a(j_a+1) + 2 j_b(j_b+1). With the twist (1, 3) this gives 0, 5, 8, 12, and with the twist (1, 1) on the representation 4 it gives 4. The slice operator reproduces these (Result 28.5).

Proposition 28.4 (the compression). Let s_v(g) = F lambda(g)^-1 v be a section in the multiplicity space of 4 on a slice, with B = F^* F, which commutes with lambda(H). Write t-hat = 2 i t_3 of Sp(1)_b. Then, with normalization by the norm of s_v,

```
Q(x_a) = (tr(B Gamma_5)/tr B) Gamma_a/5,
```

and for the functions y_T(g) = tr(T Ad(g) t-hat) with T in the Lie algebra sp(2), Q(y_T) is a multiple of T, proportional to tr(B t-hat)/tr B.

Proof. End_H(4) is spanned by 1, Gamma_5 and t-hat. Since int lambda(g) Gamma_5 lambda(g)^-1 x_a = int x_a x_b Gamma_b = Gamma_a/5, the part with Gamma_5 gives the first formula. Equivariant maps from 5 to End(4) are multiples of Gamma_a, and tr(Gamma_a Ad(g) t-hat) = 0, so 1 and t-hat contribute nothing. The second formula is Schur's lemma on the representation 10. ■

Consequently, a mode with F supported on (1, 2) and K-symmetric has tr(B Gamma_5)/tr B = +-1 and tr(B t-hat) = 0. That is the zero section: Q(x_a) = +-Gamma_a/5, and the part of a function that depends on the fibre gives zero. On the boundary with the pure level-one mode, B is the projection on (1)_{-1}, so that tr(B t-hat)/tr B = +-1. Example 16.9 has these values.

The operator and its two ends

Proposition 28.8 (endpoint restrictions). Restriction C^infinity(X, R) -> C^infinity(S^4, R) + C^infinity(CP^3, R) is surjective, and in normal coordinates a(y, v) = a(y, 0) + O(|v|), uniformly on bounded subsets of C^1(X). (Report, Proposition 1.)

Theorem 28.9 (a self-adjoint realization). Take the connection metric g = dr^2 + h(r)^2 pi^* g_{S^4} + rho(r)^2 g_{S^2} with h smooth, even and positive at 0, rho odd and smooth with rho(r) = r + O(r^3), and a product near r = l. Let Phi_M = M b(|v|^2) v . J with b smooth and positive, T_M = D_X^E + i Phi_M, and D-hat_M the doubled operator. At r = l set Z = -i c(dr), chi = Gamma-hat Z, and impose chi psi = -psi. Then D-hat_M on this H^1 domain is self-adjoint, elliptic and odd, with compact resolvent, and every smooth scalar function preserves the domain. No boundary condition enters at the zero section: after the outer condition is fixed, removing S^4 creates no further self-adjoint extension. (Report, Theorem 2, using Bär-Ballmann, Corollary 4.18.)

Proposition 28.10 (three slice operators). For the standard complex structure on CP^3: (1) the kernel of the Dolbeault-Dirac operator on O(1) is C^4 for every Hermitian metric; (2) the Levi-Civita Dirac operator twisted by the separate line bundle O(3) with its invariant connection vanishes on the 4 for every invariant squashing, which gives a kernel of dimension at least four, with any further kernel in other representations at a locally discrete set of squashings; (3) the Levi-Civita Dirac operator twisted by the full pull-back connection on E is invertible for small fibres, since (D^E_rho)^2 >= 1/(2 rho^2) - C_0 - C_1 rho^2. (Report, Proposition 3.) In Python, on the representation 4, the smallest |eigenvalue| of that operator is 3.235, 1.842, 1.200, 0.722, 0.148 at rho = 0.3, 0.5, 0.707, 1, 2. So the window comes from the competition between the normal operator and the mass.

Proposition 28.11 (normal model). Let Q = sigma . d_y + y . J on L^2(R^3; C^2 (x) C^4), with J Hermitian of spin 3/2. Then ker Q = 0 and ker Q^* = 3 (+) 1 under the normal Sp(1), with a positive gap above the zero modes; changing the sign of the mass exchanges the two kernels. The proof uses Q^* Q = -Laplacian + (y . J)^2 - sigma . J >= 0, with equality excluded since sum_i J_i^2 = 15/4, and the Callias index for the positive eigenbundles of n . J, of degrees 3 and 1. The induced bundle over S^4 is S^2(S) (+) 1, the operator on Sigma (x) S^2(S) has exactly four zero modes, in one chirality, and on the parallel inclusion S -> S (x) S^2(S) the twisted equation is a multiple of the twistor equation. So the kernel consists of harmonic spinors of the twisted operator, which are twistor spinors as sections of S. Rescaling gives the localization scale r = O(M^{-1/2}) at the core and O(M^{-1}) at the wall. (Report, Proposition 4.) In Python, in an oscillator basis: Q^* Q starts at 1.02, QQ^* has four near-zero values, with total spin 0 and three values near spin 1, then 1.02.

Lemma 28.12 (tangential mixing). On a product collar, normalized stable profiles for mass eigenvalues M q_i overlap by 2 sqrt(q_i q_j)/(q_i + q_j), independently of M, which is sqrt(3)/2 for q = 1/2, 3/2. So large mass does not remove the coupling between the stable channels O(3) and O(1), which in the pull-back connection has order 1/rho. (Report, Lemma 5.)

Proposition 28.13 (the boundary operator on the representation 4). Take the sign of Phi for which the right singular vector of T is at the zero section. Its left singular vector v is at the wall, in the domain Z v = +v of Theorem 28.9 for T^*. A wall profile exp(q(x - l)), q > 0, for T^* = i Z d/dx + D_6 + i phi tau,

with phi > 0, has q = -phi Z tau at leading order. So the stable channels of v are (Z, tau) = (+, -3/2) and (+, -1/2) twice, and they map to the channels (-, -1/2) twice. In unitary bases of these, the compression of D_6(rho) with the overlaps of Lemma 28.12 is

```
A(rho) = ( 3/(2 rho)   1/rho   rho + 1/rho )
         (     0       sqrt 2     -sqrt 2  )
```

with the matrix elements computed in Python, and reproduced by Terra 6.1 in exact symbolic arithmetic from the bracket and Clifford data. Its kernel is spanned by c = (-(2/3)(rho^2 + 2), 1, 1). The channel tau = -3/2 and one channel tau = -1/2 are supported on (1)_{-1} of the 4; the other channel tau = -1/2 is supported on (1)_{+1}. All three lie in Gamma_5 = -1, and their target spaces are mutually orthogonal, so the cross terms vanish in the traces of Proposition 28.4. Hence for the mode c

```
tr(B Gamma_5)/tr B = -1,    tr(B t-hat)/tr B = -w(rho),    w(rho) = 2(rho^2 + 2)^2 / (9 + 2(rho^2 +
↪  2)^2),
```

for every rho, with horizontal scale 1; for horizontal scale h, replace rho^2 by rho^2/h^2. So w = 8/17 as rho -> 0, w = 2/3 at the Fubini-Study radius rho = 1 (the nearly Kähler radius is 1/sqrt 2), and w -> 1 as rho -> infinity, the level-one value. The nonzero singular values of A are the next wall values on the 4: 1.863 and 2.789 at rho = 1, with the core value 2 in between. With the opposite sign of Phi, the right singular vector is at the wall, in the channels Z = -1, tau > 0, with fibre part +w(rho), and its left singular vector at the zero section.

For the connection preserving the line summands, the first row is replaced by (0, 1/rho, rho + 1/rho), the kernel is (1, 0, 0), so w = 1 for every rho, and the block on tau = -1/2 has singular values with sum of squares rho^2 + 2/rho^2 + 6 and product sqrt 2 (rho^2 + 2)/rho, so it is invertible for every rho > 0, with D^2 = 3, 6 at rho = 1. This is the collar repair of the report: an invariant connection preserving the four line summands in a collar extends into X with a cutoff, with the pull-back connection near the zero section. At the Fubini-Study metric the stable boundary model is then a sum of line-twisted Dirac operators, with four zero modes on O(3) and none on O(1), since H^q(CP^3, O(-1)) = 0 for all q.

Status. The kernel and the traces are exact for this finite block, which is the leading formal wall operator; the next correction is O(1/M) in the fixed-block expansion and does not change the limiting kernel, since the kernel is separated from the other singular values. That the boundary mode of T_M converges to it as M -> infinity is a numerical statement here (Result 28.5), since a uniform global reduction is unproved: the difference falls as phi_0 grows, from 0.037 to 0.009 at wall radius 0.5 and from 0.008 to 0.002 at wall radius near 1, between phi_0 = 16 and 32. The value 2/3 tests the factor sqrt 3/2, since without it w(1) would be 3/5.

Lemma 28.14 (a sufficient spectral criterion). Let U_M : C^8 -> Dom D-hat_M be an isometry with ||D-hat_M U_M|| <= eps_M -> 0 and ||D-hat_M psi|| >= delta ||psi|| for psi orthogonal to the range, with delta > 0 independent of M. Then for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), and the spectral space is within O(eps_M/delta) of the range. (Report, Lemma 6.) The missing steps for Conjecture 28.6 are the normal reduction with its errors, and the coercive estimate on every representation of Sp(2).

Conjecture 28.6 (the window). Take Theorem 28.9 with the sign of Proposition 28.13 and with (a) the pull-back connection and wall radius rho, or (b) the collar connection of Proposition 28.13 and the Fubini-Study metric at the wall. For large M the hypotheses of Lemma 28.14 hold, with the eight states being one copy of the 4 at the zero section in one Gamma-hat degree and one at the boundary in the other. For a family of metrics and mass profiles with a uniform positive decay rate, the off-diagonal block is exponentially small in l. The compression of a smooth function tends to (Q_B(a|*{S^4}), Q_Y(a*/{CP^3})), with Q_B and Q_Y(x_a) = -Gamma_a/5, and Q_Y(y_T) a multiple of T, equal to the level-one value of Example 16.9 times w(rho) in case (a), and equal to it in case (b). This concerns compression maps, in line with Counterexample 28.17.

Result 28.5 (Python, the radial system). We solve T on the multiplicity spaces, with Hermitian Clifford generators, so that c(dx) = i Z in the skew convention of Theorem 28.9. For horizontal scale h, after absorbing the half-density h^2 rho, T = i Z d/dx + A_0/h + A_1/rho + (rho/h^2) A_2 + i phi tau, with

-s(x) A_E/rho added for a collar connection. We take h = 1, so that T = i Z d/dx + D_6(rho(x)) + i phi(x) tau, with D_6 built from the structure constants of sp(2), rho(x) = rho_inf tanh(x/rho_inf), phi(x) = phi_0 tanh(x/w_0), w_0 = 0.3, in units with S^4 of radius 1, by the quadratic form of T^* T with P1 elements on a graded grid. This rho is not exactly a product near the wall. Rayleigh quotients of the exact form bound the values from above, and with three-point quadrature they approximate those bounds. Small entries have a mesh floor that depends on the parameters: at (phi_0, l) = (6, 4) the lowest value is 0.00548, 0.00401, 0.00355 for 2500, 5000, 10000 elements, so such entries do not establish a zero continuum value. At x = 0 the exponents are the values for R^3: +-1, +-2 (three times), +-3 on the 4.

phi_0	l = 2	l = 3	l = 4	next singular value, l = 4
3	0.406	0.159	0.066	1.293
4	0.288	0.088	0.028	1.452
5	0.200	0.045	0.011	1.665
6	0.132	0.021	floor	1.818
8	0.049	floor	floor	1.852

(rho_inf = 1, representation 4; the next value at larger phi_0 is 1.86 to 1.87.) The lowest value falls with the exponent of the constant-coefficient system, the smallest |Re kappa| of i Z (D_6(rho) - i phi tau): measured between l = 3 and 4, 0.877, 1.155, 1.464 at phi_0 = 3, 4, 5, against 0.856, 1.151, 1.476, as numerical evidence for these profiles, without a general tunnelling theorem. At large phi, kappa = phi/2 - 2/rho + O(1/phi) (phi = 64: 30.07 at rho = 1, 31.07 at rho = 2). The formula 2 mu exp(-mu l) of lemma 09 belongs to a separable interval and does not carry over.

On the two modes, at rho_inf = 1, l = 3, with the sign of Proposition 28.13, the boundary mode being the left singular vector; the core fibre parts at phi_0 = 16, 40 are below what the computation resolves:

phi_0	zero section: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B	boundary: position, Q(x_a)/(Gamma_a/5), tr(B t-hat)/tr B, Z, tau
10	0.39, -1.0000, -3e-7	2.93, -1.0000, -0.645, 0.944, -1.117
16	0.28, -1.0000, 1e-11	2.96, -1.0000, -0.658, 0.976, -1.146
40	0.16, -1.0000, 1e-11	2.99, -1.0000, -0.663, 0.996, -1.162

Against Proposition 28.13, at l = 3:

rho_inf	wall radius	-w(rho)	phi_0 = 16	phi_0 = 32	next value, phi_0 = 16	second singular value of A
0.5	0.500	-0.529	-0.493	-0.521	1.992	1.992
0.7	0.700	-0.579	-0.562	-0.575	1.966	1.966
1.0	0.995	-0.665	-0.658	-0.663	1.864	1.865
1.5	1.446	-0.788	-0.785	-0.787	1.660	1.661
2.0	1.810	-0.861	-0.858	-0.860	1.557	1.556
3.0	2.285	-0.921	-0.918	-0.919	1.487	1.486

Collar connection. With rho(l) = 1 and s(x) = (1 + tanh((x - l + 1)/0.15))/2 multiplying the vertical connection term to be removed, at phi_0 = 8, 16, 32: the boundary mode has tr(B t-hat)/tr B = -1.0000 and Q(x_a)/(Gamma_a/5) = -1.0000, the next value is 1.733, that is sqrt 3, and the core mode is unchanged. The switch is nonzero, though tiny, at x = 0, whereas the smooth extension needs a cutoff vanishing near the zero section, so these rounded values support the limiting model and do not establish exact finite-mass equality.

Other representations. The multiplicity is zero on 1, 5, 10, 14, 35. At phi_0 = 16, rho_inf = 1, l = 3, the lowest singular values are 2.188 (wall) and 2.235 (core) on the 16, 1.920 (wall) and 2.828 (core) on the 20, and 3.381 (wall) and 3.463 (core) on the 40, the core values being sqrt 5, sqrt 8 and sqrt 12 up to the mass. The multiplicity zero follows from the central character. The finite wall operator A_V(rho) has no zero at 1500 sampled radii in [0.1, 20] on the 16, 20, 40, 64, a sampled statement; its smallest singular value tends to the core value as rho -> 0 and falls slowly at large rho on the 16 and the 40 (0.33 and 0.67 at rho = 20). Higher representations were not computed, and no estimate in the representation is proved.

Core block at finite mass. The density of Proposition 28.16 is nonzero at these points, stable in the mesh to five digits:

phi_0, l	lambda	tr(B t-hat)/tr B, integrated	max of abs tr(B t-hat) over max of tr B	sup of the pointwise ratio	integrated value over lambda^2
4, 3	0.0882	-5.71e-3	6.48e-3	0.849	-0.73
5, 3	0.0453	-1.84e-3	1.97e-3	0.856	-0.89
6, 3	0.0212	-4.55e-4	4.73e-4	0.858	-1.01

The pointwise ratio reaches its maximum at the outer wall, where the core density is small, so the breaking comes from the tail of the core mode at the wall. Over phi_0 = 3 to 6, l = 2 to 4, the integrated value over lambda^2 lies between -0.4 and -1.0, an empirical observation with no O(lambda^2) asymptotic proved. These values support the hypothesis of Proposition 28.16 at the sample points, conditional on the density being nonzero in the continuum.

The compression and the algebra

Theorem 28.15 (compression, conditional on localization). Assume an eight-dimensional spectral window splitting under Gamma-hat as W_M (+) V_M with dimensions 4, 4. Then scalar compression is exactly block diagonal. If the isometries differ by at most eta_M from model embeddings with angular compressions Q_B, Q_Y and normal measures with first moments at most C M^{-1/2} at the core and C M^{-1} at the wall, then

```
|| Q_M(a) - (Q_B(a|_B) (+) Q_Y(a|_Y)) || <= C ||a||_{C^1} M^{-1/2} + 4 ||a||_infinity eta_M.
```

(Report, Theorem 7.) For a spectral wall as in lemma 24, the domain has constant boundary values, and if a|Y = *c then any boundary-localized family has* //Q{Y,M}(a) - c|| <= C M^{-1} ||d_r a||_infinity. So localization just inside a spectral wall does not recover arbitrary boundary functions for a fixed smooth a, and the limit needs a local wall or an explicitly different comparison algebra.

Proposition 28.16 (the exact symmetry test). For an equivariant copy of 4, B(r) = F(r)^* F(r) = diag(b_a(r) I_2, b_+(r), b_-(r)). All real scalar functions compress to operators commuting with the quaternionic antiunitary j if and only if b_+(r) = b_-(r) for almost every r. If in addition the 5 component is nonzero, the block generates M_2(H). If both the 5 and the 10 components are nonzero, it generates M_4(C). A vanishing integral of tr(B(r) t-hat) alone does not decide this. If a radial h has Q_M(h) = alpha I_W + beta I_V with alpha != beta, then the block projections lie in the generated algebra. (Report, Proposition 8.) At the limit, the real linear image is Herm_2(H) + Herm_4(C), of real dimension 6 + 16 = 22, and the generated real algebra has dimension 16 + 32 = 48.

Counterexample 28.17 (generation is discontinuous under convergence of compression maps). Let R_0 = Q_B E_fib, with E_fib the fibre average on CP^3, R_1 = Q_Y, and R_eps = (1 - eps) R_0 + eps R_1. These are unital completely positive, ||R_eps - R_0|| <= 2 eps, and real functions generate M_4(C) under R_eps for every eps > 0, and M_2(H) at eps = 0. So the correct limit statement concerns specified compression maps, and it gives no convergence of the generated algebras. (Report, Counterexample 9.)

Theorem 28.21 (prescribed windows on the same X). Let P_y = (1 + y_a Gamma_a)/2 and p_z <= P_{pi(z)} the tautological rank-one projector. With normalized radial profiles chi_0, chi_l supported in disjoint eps-neighbourhoods of the two ends, U_0 v = sqrt 2 chi_0 P_y v and U_l v = 2 chi_l p_z v are isometries into L^2(X; C^4) (+) L^2(X; C^4). Real functions compressed by them generate M_2(H) + M_4(C) at every eps, and the compression differs from Q_B (+) Q_Y by at most C eps ||a||_{C^1}. (Report, Theorem 13.) This proves the connected-space mechanism without Conjecture 28.6, with the window as input. Allowing nonlocal smoothing perturbations, (1 - P) D (1 - P) + P d P makes such a window reducing, which enlarges the operator class and keeps the window as input.

The finite data

Proposition 28.18 (charge conjugation and the wall). With Hermitian generators J_i, C = C_7 (x) j_E has C^2 = -1 and commutes with T_M, D-hat_M and Gamma-hat as a differential expression, and CZ = -ZC. So it exchanges the two local wall conditions and does not preserve the chosen domain. If a pointwise parent antiunitary preserves a window and commutes with real multiplication, every real compressed multiplier commutes with its restriction, which cannot hold on a generated full M_4(C). (Report, Proposition 10.) The complex boundary block therefore requires this loss of the inherited symmetry, or a different real correspondence.

Remark 28.7 (why one block is quaternionic and the other complex). Proposition 28.18 proves the signs argument of the first version, with one correction of wording: j_E commutes with the anti-Hermitian Lie algebra action and reverses the Hermitian generators, so that C commutes with i Phi. Numerically (Python, phi_0 = 16, l = 3, rho_inf = 1), exchanging the wall condition exchanges (Z, tau, tr(B t-hat)/tr B) = (0.976, -1.146, -0.658) and (-0.976, 1.146, 0.658) on the boundary mode, with the same singular values, and leaves the zero-section mode unchanged. The formal parent symmetry does not by itself prove that the finite-mass core space is preserved, since C changes the wall domain, and the breaking computed in Result 28.5 shows the size of the effect at the sample points.

Proposition 28.19 (tunnelling and the grading). If the window consists of one copy of the 4 in each Gamma-hat degree, then d_M = [[0, conj(lambda_M) I_4], [lambda_M I_4, 0]] after equivariant unitary identifications, by Schur's lemma, and lambda_M = (1/4) Tr(U_{Y,M}^* T_M U_{B,M}). The inherited grading is I_W (+) (-I_V), the auxiliary grading of the doubling, which labels the two ends under localization. The radial Clifford involution Z does not extend across the zero section, so it does not serve as an interval chirality. This grading leaves all four core states in one degree, so W = H (+) H, for the even algebra H + H + M_4(C), still needs a chosen quaternionic grading, such as a marked Gamma_p. (Report, Proposition 11.)

Proposition 28.20 (the doubled bimodule). With H = W (+) V, Theorem 16.2 gives K = H (x) conj(H), of dimension 64, with DD = d (x) 1 + Gamma-hat (x) d^c. Its odd part is exactly K^- = Hom(V, W) (+) Hom(W, V), with the desired bimodule actions, and order zero and order one hold exactly on the full K, for its actual even algebra. If lambda != 0, then DD maps K^- into K^+, P_- DD P_- = 0 and DD^2 = 2 lambda^2. So K^- is not a reducing subspace. On K^- the inherited tensor grading is constant; the sheet-sign grading of Theorem 16.7 gives J(x, y) = (y^*, x^*), J^2 = 1, J Gamma_6 = -Gamma_6 J, with zero Dirac operator. The tunnelling map W -> V does not provide the map Hom(V, W) -> Hom(W, V) of lemma 20. (Report, Proposition 12.)

Balance against Proposition 6 of lemma 24

(Report, Proposition 14, with the numerical statements updated.)

Structure	What follows
Connected scalar parent	Supplied by the chosen disc bundle X.
Independent endpoint functions	Derived, by restriction and extension.
Base-function restriction at the collapsed end	Derived geometrically; quantitative for localized states.

Structure	What follows
Four core states	Proved for the linear normal model and the resulting twisted operator on S^4.
Four boundary states	Proved for the separated collar model; for the pull-back connection, the finite block of Proposition 28.13, with the convergence numerical.
Isolated 4 + 4 window	Conjecture 28.6.
Limiting quaternionic core compression	Derived once convergence to the core model is established.
Exact finite-mass quaternionic block	Requires b_+ = b_-; numerically not satisfied at the sample points resolved.
Exact level-one boundary compression	Requires rho -> infinity or the separated collar connection.
End grading	Inherited from the chosen self-adjoint doubling.
Quaternionic grading axis	Input.
The two Hom spaces	The odd part of the coarse correspondence; the selection is additional.
KO-6 reality and order conditions	Exact after the specified correspondence and grading choices.
Nonzero connector of lemma 20	Input.
Mass value and dependence on l	Radial matching; equivariance gives only lambda I_4; the exponent is computed numerically in Result 28.5.

Choices that stay inputs: X, E = S^3(S), the hedgehog, the wall condition, the collar connection in case (b), and the doubling of Theorem 16.2. A double with two identical collapsed zero sections removes the outer boundary condition, but the same local core construction gives two quaternionic endpoint maps, so obtaining a complex block needs different structure at one end. T0 stays open, and this is a candidate for its first half.

Direction. The next analytic step is the coercive estimate of Lemma 28.14, in case (b) first, where both end models have exact kernels. Since Phi commutes with the Clifford symbol, D + iM Phi is a concentrating pair in the sense of Maridakis (arXiv:1510.07004, Definition 2.1 and Lemma 2.2), with [D, Phi] of order zero; the symbol condition goes back to Prokhorenkov and Richardson (arXiv:math/0307251). Two things are missing there. The spectral separation theorem of arXiv:2110.11654 (section 2, Assumption 3) needs a leading endomorphism independent of the normal direction, and this fails here, since (n . J)^2 depends on n for spin 3/2; so a separate core analysis or an anisotropic extension is needed. And the wall adds a boundary layer: on a half-line ||(d_t + M) f||^2 = ||f'||^2 + M^2 ||f||^2 - M |f(0)|^2, and exp(-Mt) is a zero mode away from the zero set of the mass. (Read by Terra 6.1 in arXiv HTML; the LaTeX sources were not available to it.)

Numerical check

By Claude Opus 5.5, in Python (numpy, scipy), 3 Nov 2026, in a script that ran the tests below as assertions and printed the tables of Result 28.5; by the rule of the project the script is not kept (removed on 4 Nov 2026; Terra 6.1 had reproduced its matrix of Proposition 28.13 in exact arithmetic). The slice operator on each multiplicity space is Hermitian, odd under Z, and closes on Hom_H. The structure constants reproduce [t_1, t_2] = t_3 in Sp(1)_a and Sp(1)_b. The decomposition D_6(rho) = A_0 + A_1/rho + A_2 rho holds. The form uses three-point Gauss quadrature on each element. The exact tests are {A_0, A_1} = 0, A_0^2 = 0, 4 on the 4, 5, 9 on the 16, 0, 8 on the 20, 12, 16 on the 40 (the value 0 on the 20 has the twist (1, 5) and gives no core zero mode), A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha, and at rho = 1 with the line connection the values D^2 = 0, 5, 5, 12 on the summand O(3) of the 4, 16, 20, 40, against j(j + 4) for (0, 0)-forms in O(1); the whole summand also has the value 10 on the 20. The closed form of

Proposition 28.13 is tested to ten digits, and the core mode density in the mesh.

Part 29

The eight-state window on the disc bundle

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T0, without an Astra job. It continues lemma 28 and answers the prompt 29_window_coercive_estimate.md. Refereed by Terra 6.1 (report): case (b) stands with amendments, and the two errors it found in case (a), the definition of A_eff and the sign of the fibre part, are corrected below. The referee ran no computation. A fresh reading of the corrected case (a) in the referee report on lemma 30 found one more sign and one missing step in Proposition 29.10; these are corrected, and the corrections have not been read again.

Outcome

1. **Conjecture 28.6 holds in case (b)** for the class of metrics and mass profiles of Setting 29.0: for large M, D-hat_M has exactly eight eigenvalues in (-delta, delta), for every delta < sqrt 3 (Theorem 29.7).
2. **Conjecture 28.6 holds in case (a) at the Fubini-Study radius** rho = 1, with fibre part -2/3 (Theorem 29.11). At another wall radius, case (a) follows from one kernel condition K(rho), which is a finite check at each radius.
3. **The off-diagonal block is exponentially small**, |lambda_M| <= C exp(-(1 - eta)(M/2) int f dr), by an exact tunnelling identity and Agmon estimates (Proposition 29.8).
4. **The compression converges** to (Q_B(a|*{S^4}), Q_Y(a|*{CP^3})) with error O(M^{-1/2}) (Theorem 29.9). The statement concerns compression maps, in line with Counterexample 28.17.

The proof is short because Sp(2)-equivariance gives exact sums of squares at the core and at the wall. Each estimate holds on all of L^2 of a fixed manifold, with the boundary operators elliptic on CP^3 or S^4, so uniformity over the representations of Sp(2) needs no separate argument. The spectral separation hypothesis of Maridakis, which fails for this mass (reviews/2026-11-03_terra_check_28.md, item 6), is never used. General forms of two identities are in the literature: the square with its zero-order cross term in the bulk (Maridakis, arXiv:1510.07004, Lemma 2.2), and a horizontal and vertical factorization with a curvature contraction (Kaad and van Suijlekom, arXiv:1710.03182, Proposition 18, under other hypotheses).

Scope. These are spectral statements for a chosen metric family, with flat fibres near the zero section. They give the finite window from the operator, for the sign of Proposition 28.13. They leave open: other metrics in the class of Theorem 28.9, including the Bryant-Salamon metric; the connector of Proposition 28.20; the quaternionic grading of Proposition 28.19; any dynamics selecting M, l or rho. Finite-mass generated algebras stay outside the statement (Counterexample 28.17).

Setting

Setting 29.0. Let X be the disc bundle of lemma 28, G = Sp(2), K = Sp(1)_a x Sp(1)_b, H = Sp(1)_a x U(1)_b, E = S^3(S), Phi_M = eps M f(r) n . J, and T_M = D_X^E + i Phi_M, with D-hat_M the doubled operator and the wall condition of Theorem 28.9: Z u = -u for T_M, and Z v = +v for T_M^*. Assume:

- (H1) g = dr^2 + pi^* g_{S^4} + rho(r)^2 g_{S^2}, with S^4 of radius 1, rho(r) = r on [0, 2e], and rho constant, equal to rho_l, on the collar [l - 2a, l]. Case (b) and Theorem 29.11 take rho_l = 1, the Fubini-Study metric.
- (H2) f(r) = r on [0, 2e], f = f_0 > 0 on the collar, and f > 0 on (0, l]. This is M b(|v|^2) v . J with b positive.
- (H3) The connection on E is the pull-back connection on r <= 2e. In case (b) it preserves the four line summands on the collar, with any invariant interpolation in between. In case (a) it is the pull-back connection throughout. We use radial parallel gauge: nabla_{d/dr} = d/dr. An invariant

dr-component a(r) q could otherwise be added to a line-preserving connection; it is removed by radial parallel transport, which changes the displayed formulas by a phase on each line.

- (H4) eps = -gamma_b, where gamma_b is the chirality of Sigma_4 whose K-type is (1, 2). This is the sign of Proposition 28.13: its right singular vector is at the zero section. The opposite sign exchanges T_M and T_M^* throughout. In the convention of the radial computation of lemma 28, gamma_b = +1.

On the collar put q = eps n . J, with eigenvalues +-3/2, +-1/2 on the line summands, and m = M f_0. In general q = -gamma_b n . J. In the notation of Proposition 28.13 and of the radial computation of lemma 28, q = -tau, and T_M = i Z d/dx + A + i m q on the collar, with x increasing towards the wall. The line q = +3/2 is the summand O(3), and q > 0 are the stable channels of Proposition 28.13.

Let C_R = ||A_2||, the norm on Sigma_6 (x) E of the curvature contraction A_2 = -(i/8) sum C_ab^alpha c_a c_b c_alpha; it does not depend on V, and C_R = 1 in Python. Let mu_1 > 0 be the smallest positive eigenvalue of Q^* Q for Q = sigma . d_y + y . J of Proposition 28.11. It is positive since the resolvent is compact, because (y . J)^2 >= |y|^2 / 4, and ker Q = 0. Python gives 1.02.

The zero section

Proposition 29.1 (exact splitting). Identify sections over the tube r <= 2e with K-equivariant maps s : G -> F, F = Sigma_4 (x) L^2(R^3; C^2 (x) C^4), with c(e_a) = gamma_a (x) 1 and c(f_alpha) = gamma_5 (x) sigma_alpha. Let D_H = i sum gamma_a e_a, with e_a acting as left-invariant derivatives on G, and Q^ = gamma_5 sigma . d_y + eps M y . J. Then on this tube T_M = T_0 + R, with T_0 = D_H + i Q^ and R Hermitian of order zero with |R| <= C_R |y|, and

```
T_0^* T_0 = D_H^2 + Q^^* Q^,    T_0 T_0^* = D_H^2 + Q^ Q^^*,
```

with D_H^2 commuting with Q^^* Q^ and Q^ Q^^*.

Proof. The fibres are flat and totally geodesic by (H1), so the Levi-Civita connection of X differs from the connection associated with the principal K-bundle G -> S^4 only by the O'Neill tensor. Its component is the curvature of N evaluated at v. This gives the zero-order term R, which equals r A_2 in the radial form D_6 = A_0 + A_1/rho + A_2 rho, of norm at most C_R |y|. For the pull-back connection, the remaining operator is D_H + i gamma_5 sigma . d_y + i Phi_M. The operator Q^ acts on the values of s, and it is K-equivariant, since sigma . d_y and y . J are invariant under the simultaneous rotation of y, of C^2 and of C^4. So Q^ maps equivariant maps to equivariant maps, and it commutes with each derivative e_a, because it does not depend on the point of G. Since gamma_a anticommutes with gamma_5 and commutes with y . J, D_H Q^ = Q^^* D_H, where Q^^* = -gamma_5 sigma . d_y + eps M y . J. Therefore

```
T_0^* T_0 = (D_H - i Q^^*)(D_H + i Q^) = D_H^2 + Q^^* Q^ + i (D_H Q^ - Q^^* D_H) = D_H^2 + Q^^* Q^,
```

and likewise for T_0 T_0^*. Finally D_H Q^^* Q^ = Q^ Q^^* D_H, so D_H^2 commutes with both. ■

In radial form. T_0 = i Z d/dx + A_0 + A_1/rho + i phi q, and the cross terms vanish by {A_0, Z} = 0, [A_0, tau] = 0 and {A_0, A_1} = 0. these three identities were asserted on the representations 4, 16, 20, 40 (Python, script not kept).

Proposition 29.2 (the model spectrum). On the total space of N, with f = r and T_0 as above, let Pi be the projection onto ker T_0. Then:

1. ker T_0 is one copy of the 4 of G; ker T_0^* = 0.
2. ||T_0 u||^2 >= min(4, mu_1 M) ||u||^2 for u orthogonal to ker T_0, and ||T_0^* v||^2 >= min(4, mu_1 M) ||v||^2 for all v in the domain.

Proof. Write Q^_gamma for Q^ on chirality gamma. By Proposition 28.11, Q^_gamma has kernel exactly when gamma = -eps; this is the rank-four K-representation W_0 = (1, 3) + (1, 1), and on the orthogonal complement Q^^* Q^ >= mu_1 M, after the rescaling y -> M^{-1/2} y. The same bound holds on the other chirality, and for Q^ Q^^* with the chiralities exchanged, since Q^* Q and Q Q^* have the same nonzero spectrum. The fibrewise projection P_W onto W_0, on both chiralities of Sigma_4, commutes with D_H, by Proposition 29.1. (The projection onto the fibre kernel of Q^ also selects gamma = -eps, so it does not

commute with D_H, which is odd; it commutes with D_H^2.) On chirality -eps, Q^ vanishes on the range of P_W and is bounded below by sqrt(mu_1 M) on its complement; on chirality +eps, it is bounded below by sqrt(mu_1 M) throughout. On Sigma^gamma (x) W_0, D_H is the Dirac operator of S^4 twisted by the homogeneous bundle of W_0, so by Parthasarathy's formula D_H^2 = Cas_G(V) + 3/2 - Cas_K(tau), with Cas_K = 2 j (j + 1) on Sp(1)_b, that is 0 and 4 on W_0. Only V odd under -1 occur, with Cas_G(V) >= 5/2 and Cas_G(V) >= 15/2 for V not the 4. So D_H^2 >= 4 except for D_H^2 = 0 at V = 4, twist j = 1. This kernel needs Hom_K(4, Sigma^gamma (x) (1, 3)) != 0, which holds exactly for Sigma^gamma = (1, 2), where (1, 2) (x) (1, 3) contains (1, 2) once. By (H4), gamma_b = -eps, so T_0 sees this chirality and T_0^* the other. Decomposing u by chirality and by P_W, and using the sum of squares, gives both bounds. The computation holds on compactly supported smooth sections, and extends by closure, since the doubled model is essentially self-adjoint on that core; no boundary condition enters at infinity. ■

Lemma 29.3 (core estimate). Let u be in H^1 with support in r < 2e. Then

```
|| |y| u || <= (2/M) ||T_0 u|| + sqrt(10/M) ||u||,
```

and the same holds with T_0^*. So ||T_M u|| >= (1 - 2 C_R/M) ||T_0 u|| - C_R sqrt(10/M) ||u||, and likewise for T_M^* with T_0^*. In particular, for such u with Pi u = 0, and for all such v,

```
||T_M u|| >= (2 - o(1)) ||u||,    ||T_M^* v|| >= (2 - o(1)) ||v||,    M -> infinity.
```

Proof. Q^^* Q^ = -Laplacian_y + M^2 (y . J)^2 - eps M gamma_5 sigma . J. The eigenvalues of sigma . J are 3/2 and -5/2, and (y . J)^2 >= |y|^2/4. Hence M^2 || |y| u ||^2 / 4 <= ||Q^ u||^2 + (5/2) M ||u||^2 <= ||T_0 u||^2 + (5/2) M ||u||^2, by Proposition 29.1. The rest is the triangle inequality with |R| <= C_R |y|, and Proposition 29.2. ■

The bulk

Lemma 29.4 (bulk estimate). Let w be in H^1 with support in a compact subset of 0 < r < l, and let f_1 be the minimum of f on that support. Then

```
||T_M w||^2, ||T_M^* w||^2 >= (M^2 f_1^2 / 4 - C_Phi M) ||w||^2,
```

with C_Phi the supremum of the operator norm of the Clifford contraction sum_i c(e_i) nabla_{e_i} (f n . J). With 0 < r_1 < r_2 < l and a radial cutoff psi equal to 1 on [r_1, r_2], with compact support in 0 < r < l, we have for u in Dom T_M,

```
||u||^2_{L^2(r_1 <= r <= r_2)} <= C M^{-2} (||T_M u||^2 + ||u||^2).
```

Proof. ||(D + i Phi) w||^2 = ||D w||^2 + ||Phi w||^2 + <w, i [D, Phi] w>, with [D, Phi] of order zero and pointwise bounded independently of V, and Phi^2 >= M^2 f^2 / 4. For the second statement, T_M (psi u) = psi T_M u + [D, psi] u. ■

The interpolation of the connection in (H3) takes place in this region, where it enters only through C_Phi.

The wall

Lemma 29.5 (line-twisted operators at the Fubini-Study metric). Let D_k be the Dirac operator on CP^3 twisted by O(k) with its invariant connection, k odd, at rho = 1. Then D_k^2 = C_2(L) + 3/2 - (3/8) k^2 on the SU(4)-type L, with C_2 normalized by C_2(4) = 15/8. Consequently:

1. On O(3): ker D_3 is one copy of the 4 of Sp(2), in Z = +1; and D_3^2 >= 5 on the V-isotypic part for V not the 4.
2. On O(1): ker D_1 = 0; D_1^2 >= 3 on the 4 and D_1^2 >= 6 on the V-isotypic part for V not the 4.

So the smallest nonzero |eigenvalue| on O(3) + O(1), over all V, is kappa_w = sqrt 3, attained on the 4; the lowest wall value on the 16 is sqrt 5.

Proof. The form of D_k^2 is Parthasarathy's formula for SU(4)/U(3), since the twist is a character of U(3). The Sp(2)-invariant connection on a line summand is unique, because the isotropy representation has no

trivial summand, so it is the Chern connection. The normalization: j(j + m + 3) is the eigenvalue of 2 box on (0,0)-forms in O(m) on the type [m + j, 0, j], with C_2([m + j, 0, j]) - C_2([m, 0, 0]) = j (j + m + 3), which fixes the coefficient of C_2; scalar curvature 12 gives the constant 3/2; and the term in k^2 is fixed by the kernel of D_3 on the 4. Two checks: D_0^2 = 4 on the type 6, the Kirchberg limit R/3 at R = 12 (Ann. Global Anal. Geom. 4 (1986) 291-325); and the values in the Python table below. The kernel statements are H^q(CP^3, O(1)) = C^4 at q = 0 only, and H^q(CP^3, O(-1)) = 0, with K^{1/2} = O(-2). The types of odd 4-ality with C_2 < 55/8 are the 4, its conjugate, the 20 and its conjugate (enumerated in Python). The 4 occurs on O(3) once, by the multiplicity on the slice, and this copy is the kernel; since each 20 restricts to Sp(2) as 4 + 16, neither 20 occurs on O(3). So V != 4 has C_2 >= 55/8 on O(3), giving D_3^2 >= 5. On O(1), the 4 and its conjugate restrict only to the 4 of Sp(2), so V != 4 has C_2 >= 39/8, giving D_1^2 >= 6; and the slice block on the 4 has D_1^2 = 3, 6. The chirality of ker D_3 is that of (0,0)-forms, and the slice block fixes it as Z = +1 for q = +3/2 (Python). ■

Lemma 29.6 (wall estimate in case (b)). Let w be in H^1 with support in the collar, vanishing near its inner end, and p(x) = sqrt(3 m) exp((3/2) m (x - l)), so that the kernel of T_M^* on the half-cylinder is p (x) ker D_3. Then:

1. If Z w = -w at r = l, then ||T_M w||^2 >= min(m^2/4, 3) ||w||^2.
2. If Z w = +w at r = l, and w is orthogonal to p (x) ker D_3, then ||T_M^* w||^2 >= min(m^2/4, 3) ||w||^2.

Proof. On the collar, A commutes with q and with d/dx, and anticommutes with Z. Writing w = w_+ + w_- by Z, integration by parts gives, for T_M,

```
||T_M w||^2 = ||(d + m q) w_+||^2 + ||A w_+||^2 + ||(d - m q) w_-||^2 + ||A w_-||^2,
```

where the cross terms are purely imaginary, using w_+(l) = 0. For T_M^* the same holds with q replaced by -q and with w_-(l) = 0. There are three one-dimensional cases on (-infinity, l]. (i) If g(l) = 0, then ||(d +- m q) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2. (ii) With g(l) free, ||(d + m |q|) g||^2 = ||g'||^2 + m^2 q^2 ||g||^2 + m |q| |g(l)|^2. (iii) With g(l) free, ||(d - m |q|) g||^2 is the form of -d^2 + m^2 q^2 with a Robin condition at l, whose spectrum consists of 0, on the profile sqrt(2 m |q|) exp(m |q| (x - l)), and of [m^2 q^2, infinity). Cases (i) and (ii) give the lower bound m^2/4. Case (iii) arises exactly in the channels (Z, q) = (-, q > 0) for T_M and (+, q > 0) for T_M^*. On those, expand in eigenvectors of A^2: the term ||A w||^2 is at least 3 ||w||^2, except on ker D_3 in Z = +1, which by Lemma 29.5 only arises for T_M^* at q = 3/2, where the Robin gap (9/4) m^2 applies. ■

The window

Theorem 29.7 (the window, case (b)). Under Setting 29.0 with rho_l = 1 and connection (b), let delta < sqrt 3. For M large, D-hat_M has exactly eight eigenvalues in (-delta, delta). The spectral subspace splits under Gamma-hat as W_M (+) V_M, each a copy of the 4 of Sp(2), with W_M within O(M^{-1/2}) of the cut-off core modes in the domain of T_M, and V_M within O(M^{-1/2}) of the cut-off wall modes p (x) ker D_3 in the domain of T_M^*.

Proof. We verify Lemma 28.14. Take radial functions chi_c, chi_b, chi_w with chi_c^2 + chi_b^2 + chi_w^2 = 1, chi_c = 1 on r <= e with support in r < 2e, chi_w = 1 on [l - a, l] with support in the collar. Let U_M span (chi_c u_i, 0) and (0, chi_w v_k), where u_i is a basis of ker T_0 and v_k of p (x) ker D_3, with Gram-Schmidt. Radial functions preserve the wall condition.

Trial states. T_M (chi_c u_i) = [D, chi_c] u_i + R chi_c u_i, of norm O(M^{-1/2}), since u_i has Gaussian decay at the scale M^{-1/2}; and T_M^* (chi_w v_k) = [D, chi_w] v_k, which is O(exp(-(3/2) m a)). So eps_M = O(M^{-1/2}).

Coercivity. Since [D, chi] = c(d chi) with c(d chi)^2 = |d chi|^2, and sum chi_i d chi_i = 0,

```
||T_M u||^2 = sum_i ||T_M chi_i u||^2 - sum_i || |d chi_i| u ||^2,
```

and likewise for T_M^*. The functions d chi_i vanish outside the bulk region e <= r <= l - a. So by Lemma 29.4,

(1 + C M^{-2}) ||T_M u||^2 >= ||T_M chi_c u||^2 + ||T_M chi_w u||^2 - C M^{-2} ||u||^2, ||chi_b u||^2 <= C M^{-2} (||T_M u||^2 + ||u||^2).

If u is orthogonal to every chi_c u_i, then Pi (chi_c u) = 0 exactly, since <u_i, chi_c u> = <chi_c u_i, u>, so Lemmas 29.3 and 29.6 (1) give ||T_M u||^2 >= (3 - o(1)) ||u||^2. If v is orthogonal to every chi_w v_k, then chi_w v is orthogonal to p (x) ker D_3, so Lemmas 29.3 and 29.6 (2) give the same for T_M^*. Since ||D-hat (u, v)||^2 = ||T_M u||^2 + ||T_M^* v||^2, Lemma 28.14 gives the count and the distance to the range. The window is G-invariant and close to the range of U_M, so as representations W_M and V_M are copies of the 4. ■

Remark. The gap is set by the wall at sqrt 3, with the core next at 2. On the 16, 20 and 40 the lowest values are sqrt 5, sqrt 5, sqrt 12.

Proposition 29.8 (tunnelling). Let T_M u = lambda v and T_M^* v = lambda u be a singular pair of the window, with unit vectors. By Proposition 28.19, the four singular values coincide. For a radial chi equal to 1 near r = 0 and 0 near r = l,

lambda (<u, chi u> - <v, chi v>) = <v, c(d chi) u>.

For every eta in (0, 1) there are M_eta and C_eta with

|lambda_M| <= C_eta exp(-(1 - eta)(M/2) int_e^{l - a} f dr), M >= M_eta.

Proof. The function chi u lies in Dom T_M, since chi vanishes near the wall. So <T_M^* v, chi u> = <v, T_M chi u>, and T_M chi u = chi lambda v + c(d chi) u. This gives the identity.

For the decay of u, let xi = psi exp(g) be radial, with psi = 0 on r <= e/2 and psi = 1 on r >= e, with g = 0 on r <= e, g' = (1 - eta)(M/2) f on [e, r_*], and g constant on r >= r_*. Since xi^2 u lies in Dom T_M,

||T_M (xi u)||^2 = lambda^2 ||xi u||^2 + || |d xi| u ||^2.

As in the proof of Theorem 29.7, with no core piece, the bulk bound of Lemma 29.4 and Lemma 29.6 (1), or Proposition 29.10 in case (a), bound the left side from below by the integral of (M^2 f^2/4 - C M) |xi u|^2 on the bulk, plus (3 - o(1)) ||xi u||^2 on the collar, minus C M^{-2} ||xi u||^2. On the bulk the excess over (g')^2 is at least (eta/4)(M^2 f^2 / 4) for M large, with f >= f_1 > 0, so ||exp(g) u||*{L^2(r >= e)}* <= *C_eta. For v we use the same argument from the wall inward, with g-tilde = 0 on [l - a, l], with -g-tilde' = (1 - eta)(M/2) f, and with Lemma 29.3 replacing Lemma 29.6 on the core, since T_M^* has no low states there. Take chi with |d chi| <= 2/s, supported on [r_, r_ + s] in the bulk. By Theorem 29.7, <u, chi u> -> 1 and <v, chi v> -> 0, so the identity gives |lambda| <= (4/s)* ||u||*{[r_, r_ + s]}* ||v||*{[r_, r_ + s]}. Run the weights with eta' = eta/2, and take s so small that (1 - eta')(I - int*{r_}^{r_ + s} f) >= (1 - eta) I, with I = int_e^{l - a} f. This gives the stated exponent. The absorption uses lambda -> 0, supplied by the trial states. The factor 1/2 holds on every channel, since (n . J)^2 >= 1/4. ■

For a family with l -> infinity, M >= M_eta, with uniform bounds on the end geometry, the cutoffs, the connection and the positive bulk mass, the constants are uniform, so lambda is exponentially small in l at the rate (1 - eta)(M/2) inf f. This is the uniform decay rate required by Conjecture 28.6, as amended by the referee.

Theorem 29.9 (compression). Under Theorem 29.7 or Theorem 29.11, for a in C^1(X),

|| Q_M(a) - (Q_B(a|_{S^4}) (+) Q_Y(a|_{CP^3})) || <= C ||a||_{C^1} M^{-1/2},

where Q_B(a) = int a(x) 2 P_x d mu(x), with P_x = (1 - x . Gamma)/2, and Q_Y(a) = int a(z) B(z) d mu(z). In case (b), B(z) = 4 p_z, the level-one map of Example 16.9. In case (a) at radius rho, B(z) = 2((1 + w) p_z + (1 - w) p_{jz}), with p_{jz} the quaternionic partner of p_z and w = w(rho); so Q_Y(x_a) = -Gamma_a/5, Q_Y(y_T) is w(rho) times the level-one value, and the signed fibre part is -w(rho). Restriction to the two ends is surjective (Proposition 28.8), so real functions on X generate M_2(H) + M_4(C) under the limit map, by lemma 10 and Example 16.9, for every w(rho) > 0.

Proof. Theorem 28.15, with eta_M = O(M^{-1/2}) from Theorems 29.7 and 29.11, after choosing equivariant unitary identifications of the window spaces with the model copies of the 4. The normal first moments are O(M^{-1/2}) at the core, by the Gaussian decay of the model, and O(M^{-1}) at the wall, by the profiles.

At the core, F in Hom_K(4, (1, 2) (x) (1, 3)) kills (2, 1), so B = 2 P_x in the normalization Q_B(1) = I. The sign in P_x is the value tr(B Gamma_5)/tr B = -1 of Proposition 28.13. At the wall, the pointwise inner product of different line summands vanishes, so the density is a sum over channels with weights from Proposition 28.13. ■

Case (a)

On the collar in case (a), A = A_d + A_o, where A_d = D_line preserves the lines and A_o = A_E/rho has order zero, with [A_o, q] != 0. In Python, the block of A_E between q = 3/2 and q = 1/2 has norm sqrt 3 and ||A_E|| = 2, on the full fibre, hence for all V. Let P be the projection onto {sum_{q > 0} p_q g_q : Z g_q = -g_q}, with p_q = sqrt(2 m q) exp(m q (x - l)), and Y the same with Z = +1, with Pi_Y the projection onto the second space. Let F = Pi_Y (A_d + A_o) P, from the range of P to Y, and let A_eff be the operator on the sum of both ranges with blocks F and F^*. Identified with sections over CP^3 of the lines q > 0, it is, by Lemma 28.12,

```
A_eff = D_3(rho) (+) D_1(rho) + (sqrt 3 / 2) B(rho),
```

with B(rho) the block of A_o between the stable channels, of norm sqrt 3 / rho. This is self-adjoint and elliptic on the fixed manifold CP^3, so its nonzero spectrum has a gap kappa_a(rho) > 0, uniformly in V. On the 4, F^* is the matrix A(rho) of Proposition 28.13, from Y to the range of P.

Condition K(rho). The kernel of A_eff(rho) is exactly one copy of the 4, in Z = +1, so F is injective and ker F^* is that copy.

Proposition 29.10 (wall estimate in case (a)). Assume K(rho). For w as in Lemma 29.6, with the kernel p (x) ker D_3 replaced by the kernel of the half-cylinder operator T_M^*, the two statements of Lemma 29.6 hold with the constant 3 replaced by kappa_a(rho)^2 - O(1/m). The half-cylinder kernel is exactly one copy of the 4 for m large, with fibre part tending to -w(rho) at rate O(1/m).

Proof. T_d = i Z d/dx + A_d + i m q preserves the block structure: T_d P = Pi_Y T_d P, since T_d (p_q g) = p_q A_d g. Also T_d maps the complement of P to the complement of Y, because for y in Y the function T_d^* y = A_d y lies in P. On the complement of P, ||T_d w_2|| >= (m/2) ||w_2||, by the sum of squares of Lemma 29.6, where the channels with Robin kernel are removed. So with w = w_1 + w_2 and w_1 = P w,

```
||T_M w|| >= ||(1 - Pi_Y) T_M w|| >= (m/2) ||w_2|| - ||A_o|| ||w||,
||T_M w|| >= ||Pi_Y T_M w|| >= ||F w_1|| - ||A_o|| ||w_2||.
```

Eliminating ||w_2||, one gets (1 + 2 ||A_o|| / m) ||T_M w|| >= ||F w_1|| - (2 ||A_o||^2 / m) ||w||. Under K(rho), F is bounded below by kappa_a(rho), and F^* is bounded below on the complement of its kernel. The same elimination holds for T_M^*, with P and Y and F and F^* exchanged. On the half-cylinder T_M is thus injective with closed range, so T_M^* is surjective; its kernel is the kernel of the Schur complement of the block T_d + (1 - Pi_Y) A_o (1 - P), whose inverse has norm at most (m/2 - ||A_o||)^-1. So this Schur complement differs from F^* by at most ||A_o||^2 / (m/2 - ||A_o||), and the graph of the Schur solution gives ||P_{ker T_M^*} - P_{ker F^*}|| = O(1/m). This projection estimate turns orthogonality to the exact half-cylinder kernel into orthogonality to ker F^* up to O(1/m), which gives the adjoint bound. Off the 4, this is invertible. On the 4 the block is a 2 x 3 matrix of rank 2, so its kernel has dimension exactly 1, within O(1/m) of (-(2/3)(rho^2 + 2), 1, 1). ■

Theorem 29.11 (the window in case (a)). Under Setting 29.0 with the pull-back connection, K(1) holds, and for delta < min(2, kappa_a(1)) and M large the conclusion of Theorem 29.7 holds at the Fubini-Study radius, with the wall modes replaced by the kernel of Proposition 29.10, and with fibre part tending to -w(1) = -2/3. Here kappa_a(1) >= sqrt 5 - 3/2 off the 4, and it equals 1.863 on the 4, the squared values there being (45 +- 3 sqrt 33)/8. At another wall radius the same conclusion holds under K(rho).

Proof. By Lemma 29.5, on V != 4 we have |D_3 (+) D_1| >= sqrt 5, while ||(sqrt 3/2) B(1)|| = 3/2. So A_eff is invertible there, with lower bound sqrt 5 - 3/2 > 0. On the 4, F^* is A(1), with kernel of dimension one in Z = +1, the vector (-2, 1, 1), and nonzero singular values 1.863 and 2.789, by Proposition 28.13. So

K(1) holds. The proof of Theorem 29.7 then applies with Proposition 29.10 in place of Lemma 29.6. The trial wall states are cut-offs of the exact half-cylinder kernel, so their defect is exponentially small. ■

Other radii. For K(rho), only finitely many V can carry a kernel, those with |D_3(rho) (+) D_1(rho)| <= 3/(2 rho), by ellipticity. In Python, on the 16, 20, 40 and for rho in [0.2, 4], the compressed block has smallest singular value at least 1.30. A check of K(rho) on all relevant V at a given radius needs a lower bound for D_line(rho) on all V, which we have not written down away from rho = 1.

Remarks

Remark 29.12 (the index). The summands of E on CP^3 have indices 4, 0, 0, -4, of total zero, as equivariant cobordism invariance requires: E extends over X (Witten 1983, preprint pp. 43-44). The window has one copy of the 4 in each Gamma-hat degree, so its Sp(2)-index is 4 - 4 = 0. So index theory allows lambda != 0, and the eight states come from the estimates above. The sign in (H4) is consistent with this: with the opposite sign, the end-model modes at the core belong to T_M^* and those on O(-3) at the wall to T_M (Proposition 28.13), so the two ends never lie in the same Gamma-hat degree. At finite mass the full operator can have nonzero lambda, so these states need not lie in exact kernels. The relative sign of the two ends enters through the chirality of ker D_3. It follows from the slice weights, which contain the 4 once in Z = +1 and not in Z = -1, and the sign of the core is the chirality statement behind (H4). If the O(3) zero mode lay in Z = -1 with q > 0, it would be a wall zero mode of T_M, in conflict with Lemma 29.6 (1). As a check in Python, the mean position of the modes is 0.51 at the core for T_M and 2.96 at the wall for the adjoint problem, with l = 3.

Remark 29.13 (physical units). All statements are in shape units. Under the dilation g -> R^2 g, with D -> D/R and the potential divided by R, the operator is unitarily equivalent to D-hat_M / R. So the light mass is lambda_M / R, the gap delta / R, and their ratio is unchanged, while the compression maps stay the same; this observation is from the audit of 3 November 2026 by Astra 6, section 5.2, conditional there on the estimates proved here. A family with R -> infinity, with lambda_M / delta small, closes the physical gap and keeps the eight states separated. Whether a dynamics selects such a family is open. The limiting geometry of such a family is a separate question: nothing here identifies it with a complete Bryant-Salamon manifold.

Remark 29.14 (other metrics). In the class of Theorem 28.9, T_M - T_0 also contains first-order terms with coefficients O(r^2), and M (b(r^2) - b(0)) y . J. One route shrinks the core piece to r < M^{-1/2 + s}, where Lemma 29.4 controls the complement since M^2 r^2 dominates. We have not carried this out, and the horizontal first-order terms need a bound in terms of D_H that takes account of the unbounded curvature terms.

Numerical check

Python (numpy, scipy), scripts not kept by the rule of the project. The exact inputs were asserted: [A, q] = 0 and {A, Z} = 0 at rho = 1 with the line-preserving connection, the values of Lemma 29.5 against the formula, the norms used above, and the identities of Proposition 29.1. The direct solve is a test of the prediction: the limit values below were fixed by Proposition 29.2 and Lemma 29.5 before the run, and a failure would have refuted the argument. Setting: rho(x) = x for x <= 0.4, rho(x) = 1 for x >= 1.2, phi = M rho, l = 3, with the connection switched on [0.6, 1.2], and P1 elements at 2500 cells. Lowest singular values of T_M:

V	M = 8	M = 16	M = 32	predicted limit
4	small, 1.7321, 1.9861, 2.4495	small, 1.7321, 1.9963, 2.4495	small, 1.7323, 1.9991, 2.4497	0, sqrt 3, 2, sqrt 6
16	2.2245, 2.2361, 2.4495	2.2324, 2.2367, 2.4495	2.2351, 2.2409, 2.4497	sqrt 5 (core), sqrt 5 (wall), sqrt 6
20	2.2361, 2.8148, 2.8284	2.2367, 2.8248, 2.8284	2.2409, 2.8275, 2.8286	sqrt 5 (wall), sqrt 8 (core), sqrt 8 (wall)

V	M = 8	M = 16	M = 32	predicted limit
40	3.4461, 3.4641, 3.4641	3.4585, 3.4641, 3.4645	3.4626, 3.4642, 3.4672	sqrt 12 three times

The wall values have a mesh error growing with M, of size 0.005 at M = 32 on the 16 and 20. The small value on the 4 is uninformative in this table: it halves when the mesh doubles, so it is a mesh floor, and Proposition 29.8 is proved without it. The wall mode of the adjoint problem has fibre part -1.00000 and Gamma_5 part -1.00000 at M = 8, 16, 32, as Theorem 29.9 says in case (b).

Part 30

The massless limit as a joint limit: a finite window and an opening tower

Written by Claude Opus 5.5 on 4 November 2026, for STRATEGY T14. Theorem 30.1, Corollary 30.2, Proposition 30.3, Corollary 30.4 and Proposition 30.6 were proposed in an audit of commit b2ddf98 by Astra 6 (assessment in the review); the proofs below were redone against the source lemmas. Theorem 30.5 and Proposition 30.7 are new here and rest on lemma 29, whose referee is queued. Refereed by Terra 6.1, together with the note (report); its corrections are applied and have not been read again, since the Codex quota is spent.

Outcome

The claim that W and Z become massless as a dimension opens needs a family. Along a general path to M_W = M_Z = 0 nothing forces a tower to open: lemma 23 has theta -> 0 at fixed l, with the tower spacing fixed. The families below have the hierarchy

```
m_weak << E_eps << Delta_KK -> 0,
```

so a moving low-energy window keeps a fixed finite number of states. The hierarchy alone does not make the count at fixed energy diverge: the signed spectrum 0, +-eps^alpha, +-1, +-2, … has the hierarchy with a count bounded at fixed energy below 1. So the divergence at fixed positive energy is a separate property, proved below for each family.

1. On the exact interval this holds with an internal dimension of one (Theorem 30.1), also with localized end modes (Proposition 30.3).
2. On the cylinder of lemma 24 and on the disc bundle of lemma 29 it holds with seven internal dimensions. On the disc bundle the window comes from the operator, with one copy of the 4 at each end, and the count at fixed energy has seven-dimensional Weyl growth (Theorem 30.5).
3. On the disc bundle the core block of the window is quaternionic up to a defect that is O(M^{-1/2}) in general, and exponentially small when the connection preserves the quaternionic structure of E, for all bounded real observables (Propositions 30.6, 30.7).

Scope. These are statements about operators, compression maps and spectra along families given by hand. No dynamics selects them (Remark 30.8). The vector masses of Corollary 30.2 belong to the five-dimensional Yang-Mills toy of lemma 23. On the disc bundle only the fermion window is treated, so the relation between lambda and M_W is unproved there.

The interval

Theorem 30.1 (window and tower on the interval). Let (H_F, D_, *J_F, Gamma_F) satisfy the hypotheses of Theorem 22.1, with the algebra A_F represented by pi as in Theorem 22.4, r = dim H_F, and for 0 < alpha < 1, l_ > 0* and small eps > 0 let D_F(eps) = eps D_, *l_eps = l_* eps^{-alpha}, with D_eps the interval operator D_X of Theorem 22.1 for these data. Let Delta_eps be the smallest element of spec |D_eps| outside spec |eps D_*|. Then for eps small:

1. W_eps^* D_eps W_eps = eps D_*, with J_F and Gamma_F exact.
2. Delta_eps = (pi / l_) *eps^alpha - eps //D_||*.
3. ||Q_eps(a) - pi(a)|| <= (2/3) l_^2 *//D_||*^2 eps^{2(1 - alpha)} ||a||.
4. For alpha < beta < 1, E_* > 0 and E_eps = E_* eps^beta, the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals r.
5. For fixed E > 0, N_eps(E) ~ (2 r l_* E / pi) eps^{-alpha}.

Proof. The condition l_eps ||D_F(eps)|| = l_* ||D_// *eps^{1 - alpha} < pi/2 holds for small eps. Theorem 22.1 gives item 1, and the next level is (pi - theta_max)/l = pi/l - //D_F//, which is item 2. Theorem 22.4 (2) gives item 3. For item 4, eps //D_||* « eps^beta « eps^alpha. For item 5, the signed spectrum is (n pi +- theta_k)/l_eps, which is r arithmetic progressions of spacing pi / l_eps, counted with multiplicity also when singular values coincide or vanish, each with (2 E l_eps / pi)(1 + o(1)) points in [-E, E]. ■

Theorem 22.1 starts from supplied finite data and a matrix-valued parent algebra, and the wall data change with eps by hand. A fixed Majorana block does not scale as eps D_*, so applying the theorem to an electroweak Dirac block needs a reducing sector or a treatment of the heavy neutral sector.

Corollary 30.2 (vector poles in the toy of lemma 23). For the universal channel of lemma 23, with m_eps = eps m_* and theta_eps = l_eps m_eps -> 0,

```
M_W = arcsin(sin(2 theta_eps)/sqrt 2) / l_eps,    M_Z = arcsin(2 sin(2 theta_eps)/sqrt 5) / l_eps,
```

so M_W^2 = 2 m_eps^2 (1 + O(theta_eps^2)) and M_Z^2 = (16/5) m_eps^2 (1 + O(theta_eps^2)). The vector poles follow the hierarchy. The constants 2 and 16/5 belong to this five-dimensional action. The small-mass pole coefficients of the fully massive finite spectral action are 3 and 24/5 (lemma 18, Proposition 14, and lemma 23), for the family with both members of every doublet at the same Dirac mass and no Majorana mass. Lemma 25 connects the two pairs at the level of the leading coefficient: its M^2 = ((3 - d_eff(s))/2) ||[D_F, T]||^2 / ||T||^2, with s = l Lambda, is the leading Taylor quotient of the heat-cutoff action for a generator constant in the transported frame, so the coefficient is 3 and 24/5 at d_eff = 0, and 2 and 16/5 at d_eff = 1. This does not identify the spectral-action pole problem with the Yang-Mills toy, and a joint-limit use needs a remainder uniform in s, since lemma 25 expands at fixed s. Along Lambda = E_eps we have s = l_* E_* eps^{beta - alpha} -> 0, the finite value, and at fixed Lambda > 0 we have s -> infinity, the interval value. Varying the cutoff of the action is an extra prescription: an observational cutoff alone does not change the action.

Proposition 30.3 (a localized family). Take M = -mu and the mixed conditions of lemma 9, with mu_* > 0, 0 < alpha < 1, mu_eps = mu_* eps^alpha and the length l_eps^loc = c_eps / mu_eps, where c_eps = (1 - alpha) log(1/eps) + c_0. This length differs from the l_eps of Theorem 30.1. Let s = kappa l be the root of s = c tanh s, and m = mu sech s. Then

```
m_eps ~ 2 mu_* exp(-c_0) eps,    Delta_eps ~ mu_* eps^alpha,    (1/kappa) / l_eps = 1/s -> 0.
```

With the left profile L(x) proportional to exp(-kappa x) - r exp(kappa x), r = exp(-2 kappa l), the following identities are exact:

```
r = (m / (mu + kappa))^2,    mu - kappa = m^2 / (mu + kappa).
```

Proof. Lemma 9.5 gives m = 2 mu exp(-c)(1 + o(1)) and kappa/mu -> 1. The first oscillatory root has pi < k_1 l < 3 pi/2 and lambda_1^2 = mu^2 + k_1^2, so Delta/mu -> 1. From mu = kappa coth s and m = kappa / sinh s we get mu + kappa = kappa exp(s) / sinh s, hence m / (mu + kappa) = exp(-s). The second identity is m^2 = mu^2 - kappa^2. ■

The localization width and the length both grow, with a shrinking ratio. The finite two-point distance 1/m and the length l are both divergent, with (1/m)/l ~ exp(c)/(2c), so a metric convergence statement would need its own comparison maps.

Corollary 30.4 (the cylinder of lemma 24). Use the operator D_r = E d_x + Z A / r on the interval times CP^3, with A as in lemma 24, finite data on V = ker A = C^4 with gradings of type (2, 2) as in lemma 24, so that r = 4, the scaling of Theorem 30.1, and r_eps = r_* eps^{-gamma}, 0 < gamma < 1. Other values of r need an enlarged tangential bundle, which we do not construct. Then for max(alpha,

gamma) < beta < 1, the cutoff E_* eps^beta isolates the finite window, since the next level is at least min((Delta_K / r_) *eps^gamma, (pi / l_)* eps^alpha - eps ||D_*||), by lemma 24, (5). At fixed E > 0 the count is proportional to l_eps r_eps^6 E^7, the seven-dimensional Weyl growth, by Dirichlet and Neumann bracketing of the squared product operator as l_eps and r_eps grow. For gamma = alpha the geometry expands homothetically in all seven directions.

The window here is prescribed by the spectral walls of lemma 24.

The disc bundle

Theorem 30.5 (joint limit on the disc bundle). Fix a shape as in Setting 29.0, with connection (b), or with connection (a) at rho = 1. Let delta > 0 be as in Theorem 29.7 or 29.11, G_0 = int_e^{l - a} f dr, 0 < alpha < beta < 1, E_* > 0, eta in (0, 1), and

```
M_eps = (2 (1 - alpha) / ((1 - eta) G_0)) log(1/eps),    R_eps = eps^{-alpha},    D_eps =
↪  D-hat_{M_eps} / R_eps.
```

By the dilation of Remark 29.13, which is exact under the canonical spinor identification and the L^2 unitary factor R^{-7/2}, D_eps is the operator of the metric R_eps^2 g with potential (M_eps / R_eps) f(s / R_eps) n . J in the physical radial coordinate s = R_eps r, of size M_eps / R_eps -> 0. For eps small:

1. D_eps has exactly eight eigenvalues in (-delta eps^alpha, delta eps^alpha), and they are +-m_eps with multiplicity 4 (or zero with multiplicity 8), where m_eps <= C eps. The next level is at most C log(1/eps) eps^alpha, so it tends to zero.
2. For E_eps = E_* eps^beta the count N_eps(E_eps) = #{|D_eps| <= E_eps} equals 8.
3. The compression converges to the endpoint maps, with error C ||a||_{C^1} (log 1/eps)^{-1/2} in shape coordinates.
4. For fixed E > 0, N_eps(E) ~ C_X E^7 eps^{-7 alpha}, with C_X = 64 omega_7 Vol_g(X) / (2 pi)^7 the Weyl constant of the elliptic operator D-hat_0 on X, where omega_7 is the volume of the unit ball and 64 = 8 x 4 x 2 counts the spinors, E and the doubling.

Proof. Item 1: Theorem 29.7 or 29.11 applies since M_eps -> infinity, Proposition 28.19 gives the multiplicity, and Proposition 29.8 gives lambda <= C exp(-(1 - eta)(M_eps/2) G_0) = C eps^{1 - alpha}, hence m_eps = lambda / R_eps <= C eps. The bound on the next level: the shape-unit value is at most s_9(D-hat_0) + c M_eps, with the comparison below. Item 2: eps « eps^beta « eps^alpha. Item 3: Theorem 29.9, with M_eps^{-1/2} of the stated order. Item 4: D-hat_M = D-hat_0 + M V on the same domain, with V bounded by c = (3/2) sup f, so by min-max N(D-hat_0, Lambda - c M) <= N(D-hat_M, Lambda) <= N(D-hat_0, Lambda + c M). Take Lambda = E R_eps, so that M_eps / Lambda -> 0, and use the Weyl law for the fixed self-adjoint elliptic operator D-hat_0. ■

So the eight finite states stay separated at the moving scale, and at fixed physical energy the spectrum has the growth of a seven-dimensional internal space, for eleven dimensions in M^4 x X. Item 1 is an upper bound on m_eps, since a lower bound on lambda is unproved. A family with l_eps growing at fixed M has the same form, with the rate of Proposition 29.8. The constants M_eta, C_eta of Proposition 29.8 are fixed once the shape and eta are fixed, so no uniformity in a varying shape is used. The limit at fixed energy resolves a neighbourhood whose shape is fixed and whose size grows, so the pointed limit is flat at a fixed interior point and a half-space at a fixed boundary point. Nothing here identifies it with a complete Bryant-Salamon manifold, or with a solution of field equations.

The quaternionic defect

For the 4-copy of the window at the core, let B(r) = F(r)^* F(r) = diag(b_a I_2, b_+, b_-) as in Proposition 28.16, let j be the quaternionic antiunitary commuting with Sp(2), and P_H(A) = (A + j A j^-1)/2. Put

```
eta_t = int |b_+(r) - b_-(r)| dr / int tr B(r) dr.
```

Proposition 30.6 (all bounded real observables). For the normalized compression Q of this mode, ||Q(f) - P_H Q(f)|| <= 2 eta_t for every real f with ||f||_infinity <= 1. If X_1, …, X_k are such compressions, then ||X_1 … X_k - P_H(X_1 … X_k)|| <= 2 k eta_t.

Proof. With angular Haar measure of total mass 1, Schur's lemma gives the angular average (tr B / 4) I, so Q(f) = (4 / int tr B) int int f(r, g) lambda(g)^-1 B(r) lambda(g) dg dr. Since j commutes with lambda(g), Q(f) - P_H Q(f) is the same integral with B - P_H B. Here j exchanges b_+ and b_- and fixes the block b_a, so ||B - P_H B|| = |b_+ - b_-|/2. The triangle inequality gives the first bound. For the second, write X-tilde = j X j^-1, so that X - P_H X = (X - X-tilde)/2 and ||X_i - X-tilde_i|| <= 4 eta_t. Telescoping, X_1 … X_k - X-tilde_1 … X-tilde_k is a sum of k terms, each a product of contractions with one factor X_i - X-tilde_i, so its norm is at most 4 k eta_t; dividing by 2 gives 2 k eta_t. ■

So bounded expressions become quaternionic when eta_t -> 0. An unrestricted generated algebra may use coefficients of size 1/eta_t and retain the other directions, as in Counterexample 28.17.

Proposition 30.7 (the size of the defect). Under Theorem 29.7 or 29.11, eta_t = O(M^{-1/2}). Assume moreover that the quaternionic structure j_E of E is parallel on all of X. This holds for the pull-back connection and for the interpolation (1 - s(r)) nabla^pull + s(r) nabla^line with real s, and it can fail for other invariant interpolations allowed by (H3). Then for every eta in (0, 1) there are M_eta and C_eta with

```
eta_t <= C_eta exp(-(1 - eta) M int_{2e}^{l - a} f dr),    M >= M_eta.
```

This is the exponent of the bound on lambda in Proposition 29.8 doubled, with the integral starting at 2e. A bound in terms of lambda itself would need a lower bound on lambda, which is open.

Proof. For the first bound, fix a normalized equivariant model F_0 at the core with C F_0 = F_0 for the local real structure below, so that j B_0 j^-1 = B_0. Theorem 29.7 gives ||u - chi_c F_0|| = O(M^{-1/2}) in L^2, and |b_+ - b_-| <= (|F| + |C F|) |F - C F| gives the bound, by the Cauchy-Schwarz inequality.

For the second, j_E is parallel and j_E (n . J) j_E^-1 = -n . J, so C = C_7 (x) j_E commutes with D_X^E and with i Phi_M, hence with T_M as a differential expression, with C^2 = -1, and it exchanges the wall conditions (Proposition 28.18). For the line-preserving connection, j_E P_q j_E^-1 = P_{-q} gives j_E nabla^line j_E^-1 = nabla^line, and real affine interpolation keeps this. On Hom_G(4, -) put (C F) = C F j^-1, which is antilinear with square +1. It commutes with D_H, with the full fibre term i Q^ and with R separately, since each part of the Clifford structure transforms in the same way, so it acts on the one-dimensional space Hom_G(4, ker T_0) as a real structure. Fix F_0 there with C F_0 = F_0, so that B_0 is j-invariant. Let u = F be the mode with T_M u = lambda v, T_M^* v = lambda u, with the phase for which <chi_c F_0, u> is real, with chi_c as in Theorem 29.7. Put zeta = u - C u and nu = v - C v. In the interior,

```
T_M zeta = lambda nu,    Pi (chi_c zeta) = 0,
```

where the second holds since <chi_c F_0, C u> is the conjugate of <chi_c F_0, u>. Take the weights of the proof of Proposition 29.8 starting at r = 2e and meeting at r_* = l - a: g = 0 on r <= 2e, g' = (1 - eta)(M/2) f on [2e, l - a], g constant beyond; and g-tilde = G - g on r <= l - a, g-tilde = 0 beyond, with G = g(l - a). So g-tilde is constant on r <= 2e. That proof gives ||exp(g) u|| <= C_eta on X, and ||exp(g-tilde) v|| <= C_eta on r <= l - a/2, and lambda <= C_eta exp(-G). The same bounds hold for C u and C v, since C is pointwise isometric.

Let xi = psi-tilde exp(g-tilde), with psi-tilde = 1 on r <= l - a and 0 on r >= l - a/2. Then ||T_M (xi zeta)|| <= lambda ||xi nu|| + || |d xi| zeta ||. Since xi is constant on the support of chi_c, Pi (chi_c xi zeta) = 0, so Lemmas 29.3 and 29.4 and the partition of Theorem 29.7 bound ||T_M (xi zeta)|| from below by the bulk term with (M/2) f and by (2 - o(1)) on the core. On the bulk |d xi| = (1 - eta)(M/2) f xi, which is absorbed with the factor eta to spare, leaving

```
||xi zeta|| <= C (lambda + ||u||_{L^2(l - a <= r <= l - a/2)}) <= C exp(-G).
```

Finally B - j B j^-1 = F^* F - (C F)^* (C F), so |b_+ - b_-| <= (|F| + |C F|) |zeta| pointwise in r. On r <= l - a, the Cauchy-Schwarz inequality with the weight exp(g) gives

```
int (|F| + |C F|) |zeta| <= 2 ||exp(g) u|| ||exp(-g) zeta|| = 2 ||exp(g) u|| exp(-G) ||xi zeta|| <=
↪  C exp(-2 G),
```

since exp(-g) = exp(g-tilde - G). On r >= l - a, both |F|^2 and |C F|^2 integrate to at most C exp(-2 G). ■

The samples in section 6.1 of the audit (Python, in the pull-back connection, rows reproduced by Claude Sonnet 5.5) have eta_t / lambda^2 between 0.5 and 0.9. They vary the length at fixed mass, with profiles outside Setting 29.0, so they are a qualitative comparison and do not test this statement as M -> infinity. Under the added hypothesis the defect comes from the wall, which breaks C: for an operator with the same interior and no wall, on a complete manifold, Hom_G(4, ker T) carries a real structure, and if it is one-dimensional the core block is exactly quaternionic.

Remark 30.8 (what is missing, and the direction). The next steps are, in order:

1. The vector sector on the family of Theorem 30.5: the kinetic matrix and mass Hessian of the vector fluctuations, H_eps u = M^2 K_eps u, and then the full momentum-dependent kernel. The aim is M_W, M_Z ~ c lambda / R with finite nonzero couplings. The tunnelling eigenvalue alone does not establish this.
2. The gauge normalization. For a bulk Yang-Mills field with fixed microscopic coupling and constant profile, g_4^-2 = Vol(X) / g_11^2, so plain dilation gives g_4 proportional to R^{-7/2}, which must be separated from restoration at finite coupling. Gauge fields from isometries have Killing overlap integrals instead.
3. The action in the radius and the electroweak order parameter. A toy V = a r^-p + b eps^2 r^q with a, b, p, q > 0 has its unique minimum at r proportional to eps^{-2/(p+q)}, so alpha = 2/(p+q) lies in (0, 1) if p + q > 2. The repository has not derived such a potential, and adding it by hand would derive nothing.
4. The connector of Proposition 28.20, and the origin of the quaternionic grading of Proposition 28.19.

Numerical check

No statement here needs a computation. The audit tabulates the localized family in Python (mu_* = 1, alpha = 1/2, c_0 = 2), and Claude Sonnet 5.5 recomputed the rows; for eps = 10^-2, c = 4.3026 and m = 2.710 x 10^-3, against the asymptotic 2.707 x 10^-3.

Part R11 (referee report, lemma 24)

Referee report on lemma 24

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_24.md. Reproduced as delivered, with links made relative. The amendments are collected at the end of lemma 24.

The main operator-window construction and the new one-loop argument are correct in their stated families. The principal gap is the statement of convergence of compression maps for general observables as the interval shrinks: the varying source spaces need an identification, and different interval profiles have different limits.

This report gives one verdict for each of the ten requested items. The checks are analytic; no numerical computation was used.

1. **Result 1 — Correct.** In Result 1, $C_i^2 = \Pi$, $(\operatorname{sgn} A)^2 = 1 - \Pi$, and the cross terms vanish. Thus both χ_i are self-adjoint involutions and anticommute with E. They commute with $\hat{\gamma}$, so the boundary domain is preserved by the grading.

   Let $B = SA/r$. At 0, the complementary boundary condition selects $B < 0$; at ℓ, it selects $B > 0$. The reversal makes the boundary term in (3) nonnegative and gives (2). The finite kernel conditions are maximal isotropic and give the interval operator of lemma 23.

   The H^1 realization is correct. Decomposition into eigenspaces of A gives self-adjoint finite-dimensional interval problems. On their direct sum, (3) controls both q' and Aq; ellipticity of A identifies the graph domain with the stated H^1 domain. Rellich compactness then gives compact resolvent. The projection Π reduces both the expression and the domain.

Differentiation verifies (4). The principal-logarithm hypothesis gives $\ell\|D_F\| < \pi/2$, and the interval spectrum together with (2) gives (5). Thus, when $\Delta_K/r > \|D_F\|$, the displayed embedding is the isolated finite window.

2. **Result 2 — Correct with amendment.** With the endpoint convention made explicit, condition (6) gives the claimed local self-adjoint elliptic realization. Write G_0 at $x = 0$ and G_ℓ at $x = \ell$, or write G_i at $x = i\ell$ for $i \in \{0, 1\}$. The present "at $x = i$" leaves the index convention implicit.

   For $\chi = ZG$, the adapted tangential principal symbol exchanges the two χ-eigenspaces because $[G, \sigma_A] = 0$. The normal symbol exchanges them as well. Bär–Ballmann, Corollary 4.18 and Theorem 4.11 give the relevant ellipticity and self-adjointness criteria. Their Example 4.20 assumes anticommutation with the full adapted operator; (6) supplies the principal-symbol condition needed for Corollary 4.18.

   Equation (7) is correct for the stated maximal smooth scalar **$*$-algebra**. Preservation by both f and $\bar{f}$ makes the boundary subspace reducing, hence gives commutation with χ_i. Commutation with $C_i^2 = \Pi$ then forces multiplication by f_i to preserve $H^0(\mathbb{CP}^3, \mathcal{O}(1))$. Those sections have no common zero, so f_i is holomorphic and therefore constant.

   The local hypothesis is restrictive. On the unaugmented Dolbeault spinor bundle, the six-dimensional complex Clifford representation is irreducible. A fibre endomorphism commuting with every $\sigma_A(y, \xi)$ is scalar; a smooth self-adjoint involution of this form is consequently $+1$ or -1 on connected K. Additional fibres and other local elliptic boundary projectors are directions outside this sufficient criterion.

3. **Result 3 — Correct with amendment.** Equations (9)–(11) follow as stated. Equation (9) proves that $g^2 = 1$ is equivalent to $[G, \Pi] = 0$. Scalar multiplication commutes with G, giving (10); the complex-linear commutant of the generated $M_4(\mathbb{C})$, or of $M_2(\mathbb{H})$ acting on $\mathbb{C}^4$, is $\mathbb{C}1$. Hence the self-adjoint involution g is ± 1.

   The auxiliary example also checks: $Q_K(y_a) = \Gamma_a^{\text{section}}/5$, and the Clifford traces give $\operatorname{tr} g^2 = 16/5$. Applying (9) and taking the largest eigenvalue gives the displayed $2/\sqrt{5}$ lower bound on boundary-trace mixing.

   Amend the marked-point sentence at lines 236–244 to say that the **natural field displayed there** supplies no separately marked point. A local field can encode a chosen point: with the auxiliary bundle, $G_p(y) = p_a \Gamma_a^{\text{aux}}$ is a local constant field whose coefficients contain p. The restriction (11) concerns the unchanged four-section window and an actual involution obtained by compression. A coupled window, extra fibres, or spectral walls remain directions to examine.

4. **Result 4 — Correct with amendment.** Inserting W_ℓ into the scalar multiplier gives (12), including its independence of r. For real S^4 functions, (15) follows from quaternionic Hermiticity and the six independent matrices $1, \Gamma_a$. When the walls are aligned,

$$\mathcal{E}_p(\Gamma_a) = p_a \Gamma_p,$$

   so direct scalar compression gives (13). Generating $M_2(\mathbb{H})$ first and then taking its Γ_p-centralizer gives (14).

   For (16), state explicitly that the two Hom spaces and the two product-factor scalar algebras of Example 16.9 have been supplied. The V-side left action on the second Hom space commutes with its right grading, giving

$$\mathbb{R} \oplus \mathbb{R} \oplus M_4(\mathbb{C}).$$

   The endpoint argument proves equality of the **finite-dimensional linear images**: interior-supported scalar functions separate the same matrix-entry densities. It does not by itself identify the parent algebras, their norms, gauge groups, or one-forms. The restriction $\mathbb{R} \oplus \mathbb{R}$ applies to real S^4 scalars, aligned walls, and direct compression. Intermediate algebra generation and other windows remain available constructions.

5. **Result 5 — Gap.** Equations (17)–(20) are correct under their displayed hypotheses. In particular, the complementary resolvent in (17) has norm at most r/Δ_K; at fixed D_F, the interval tower also decouples as $\ell \to 0$. The Schur estimates (18) hold with the stated positive g_r. Equation (19) holds for an x-independent ambient multiplier, and (20) follows from the even part of the conjugation series.

   The final assertion about convergence of **the compression maps** needs a common domain and a specified collapse map. For example, take aligned walls, $D_F = 0$, and the admissible scalar functions

$$f_\ell(y, x) = 6(x/\ell)(1 - x/\ell).$$

   Both endpoint values are zero, but $\Phi_\ell(f_\ell) = 1$. Thus a general family of observables cannot be replaced by its value at the collapsing endpoint.

   One exact correction is to fix $F(y, s)$ on $K \times [0, 1]$, set $f_\ell(y, x) = F(y, x/\ell)$, and state

$$\Phi_\ell(f_\ell) \longrightarrow \mathcal{E}_0 Q_K \left(\int_0^1 F(\,\cdot\,, s)\, ds \right).$$

   For x-independent ambient multipliers, the limit is $\mathcal{E}_0 Q_K$, as follows from (19). Nonconstant x-independent functions on K are ambient multipliers; they do not themselves obey the spectral-wall endpoint restriction. Result 4's equality of finite images supplies representatives with the required endpoint values.

   The rate discussion after (18) also needs a proof or a strengthened assumption: the two displayed rv_r conditions alone do not control the denominator g_r. A simple sufficient addition is

$$r(b_r + L) \le (1 - \varepsilon)\Delta_K$$

   for some fixed $\varepsilon > 0$. The stated $b_r = O(1/\ell)$ case can instead be justified from the stronger complementary estimate

$$|\mathcal{D}_{r,Q}| \ge \left(\frac{\Delta_K^2}{r^2} + \frac{\pi^2}{4\ell^2} \right)^{1/2},$$

   obtained by applying the one-ended Poincaré inequality to the positive and negative B-spectral components. Together with $g_r > 0$ and fixed L, this supplies a complementary gap of order $1/r$.

6. **Result 6 — Correct.** The table in Result 6 separates the chosen data from the algebraic consequences. The proposed $\widehat{J} = E\mathcal{J}_1$ has the stated signs and induces $J_F = \Gamma_0 J_1$ when all the displayed commutation and anticommutation hypotheses hold. Those are sufficient conditions; the note does not construct the required tangential extension or establish its order-zero property.

   The quaternionic j commutes with the Γ_a, so j alone fails the required anticommutation with Γ_p. At the finite level, if the two gradings are Γ_p, Γ_q, a unit $n \perp p, q$ gives $J_1 = \Gamma_n j$ with the required square and anticommutation. Extending that choice to the stated parent remains a separate task.

   The connector calculation gives the displayed algebra after choosing the two complex lines and nonzero coupling. A marked quaternionic line supplies neither of those complex lines by itself.

7. **Result 7 — Correct.** For No-go theorem 22.7, the endpoint-preserving unitary $v = u\, u(\ell)^*$ implements the fluctuation and preserves the original domain. The corresponding real unitary works under the stated order-zero condition. Consequently the moving spectral window is unitarily equivalent to the original one, with the same singular values. The displayed fixed-compression example also has the stated sign and can change singular values at second order.

   The conclusion concerns one interval, fixed wall coupling X, zero bulk mass, scalar inner fluctuations from the even algebra, and the moving spectral window. It is an exact spectral statement without an action or loop expansion. Variable wall coupling, odd connection directions, suitable bulk masses, further internal connection data, and additional fields lie outside those hypotheses. The result directs the search for a radial variable toward those constructions; it does not decide the full fluctuation space of the $K \times I_\ell$ parent.

8. **Lemma 8 and Theorem 9 — Correct with amendment.** The proof of Lemma 8 checks. The displayed formula for b'' is positive for $x > 0$, since $\sinh x \cosh x > x$. Log-sum-exp is convex and increasing in each argument, so the integrand in (21) is jointly log-concave. Its integral is finite for every u, and Prékopa's Theorem 6, p. 342 applies. Differentiation under the integral gives $0 < e < 1$ and nondecreasing e.

   There is also a separate analytic check of the key monotonicity, using a **covariance argument**. Let

$$\rho(s) = \frac{(\operatorname{arsinh}\sqrt{s})^3}{2\sqrt{s(1+s)}}, \qquad L(v) = \log \rho(e^v).$$

   The substitution $s = \sinh^2 x$ writes $\mathcal{J}(z) = \int_0^\infty \rho(s)/(s+z)\, ds$. For $v = 2\log\sinh x$,

$$L'(v) = \frac{3\tanh x}{2x} - \frac{1+\tanh^2 x}{2},$$

   which is strictly decreasing: $x\operatorname{sech}^2 x - \tanh x < 0$. With $z = e^u$, let the probability density in y be proportional to $e^y \rho(e^{u+y})/(1+e^y)$. Then

$$e(e^u) = \mathbb{E}\left[\frac{1}{1+e^y}\right], \qquad \frac{d}{du} e(e^u) = \operatorname{Cov}\left(\frac{1}{1+e^y}, L'(u+y)\right) > 0.$$

   Both functions in the covariance are strictly decreasing. This check uses the same integral but does not use the log-concavity or Prékopa argument.

   For Theorem 9, the differential inequality for R_c follows from $a = ct(1-t) \geq t$, $e(a) \geq e(t)$, and $e(t) < 1$. Dominated convergence gives

$$H(0+) = 54/5, \qquad H(1/2-) = 0.$$

   Hence H is strictly decreasing and all three conclusions follow. The independent covariance check above verifies the essential monotonicity; the final classification uses the derivative calculation (23).

   Amend the theorem to state $\theta \in [0, \pi/2]$ explicitly. At $\theta = \pi/2$, the determinant has the stated continuous endpoint value, while the isolated principal-logarithm window is degenerate. The no-interior-minimum conclusion assumes the displayed flat, massless, one-loop determinant, its gauge multiplicities and connector wall condition, frequency-one fermions, no fermion-connector modification, and no additional angle-dependent term. Modified form factors, fermion sectors, bulk masses, wall terms, and the uncomputed K-tower effective action remain directions for an interior vacuum.

9. **Result 10 — Correct with amendment.** For real finite opposite masses, identity (17) of this result's discussion—namely equation (10)'s displayed G-identity at lines 658–663—makes the fermion contribution strictly decreasing in t. With the separately assumed flat gauge terms and $n_f > 0$, $t = 1$ is the unique global minimum.

   For $a = c_Q + c_U \neq 0$, the displayed expansion of G and of the logarithm is correct. State that the divergent contribution has nonzero fermion multiplicity and is considered without cancellation by additional sectors. A regulator, subtraction and specified physical completion are needed to draw a conclusion from the general-mass potential.

   The opposite-mass Python scan in lemma 23 covers the listed finite sets of masses and multiplicities and 721 sampled angles. It gives a numerical result for those samples; it does not exclude a local minimum between samples or at unsampled parameters. General masses with a specified renormalized completion and changed gauge form factors remain open directions.

10. **Consequence for T0 and T0b, and the note — Correct with amendment.** The consequence keeps the central questions open: the construction supplies an operator and grading window with prescribed spectral walls, while the coupled local scalar window, compatible reality, second sheet and connector require further data. The no-go statements do not close either T0 or T0b.

In the note, replace "higher modes" in the radius-collapse sentence by **higher K-modes**. At fixed ℓ, the interval tower remains; its decoupling uses the subsequent $\ell \to 0$ limit. Also identify the exact window there as the operator-and-grading window, since the required parent reality structure is conditional.

At the one-loop statement, state $0 < n_f < 54/5$ explicitly for the unique interior maximum and retain the determinant assumptions listed in item 8. When this review is recorded in the repository, update the provenance sentence at line 185 to identify the review and Astra 6's one-loop proof.

Amendments and errors

- Make the local-wall endpoint indices explicit and distinguish Bär–Ballmann's principal-symbol criterion from its stronger full-operator example.
- Qualify the marked-point statement to apply to the displayed natural field.
- State the product-factor inputs for (16) and the scope of equality of compression images.
- Define the source identification and averaging map for interval-collapse convergence; add a denominator estimate to the mixing-rate proof.
- State the angle interval, endpoint degeneracy, and complete determinant hypotheses for Theorem 9 and the paper note.
- Scope the bulk-mass scan to its sampled parameters and state the nonzero-multiplicity assumption for the ultraviolet divergence.
- Update the note's radius-collapse and provenance wording when the review is recorded.

No error was established in the displayed equations. Item 5 has the stated gap in the general compression-map conclusion.

Overall: 3 Correct; 6 Correct with amendment; 1 Gap; 0 Error.

Part R15 (Terra referee, lemma 28)

Referee report on lemma 28: Proposition 28.13 and the corrected numerics

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_28.md. Reproduced as delivered, with links made relative. The corrections are applied in lemma 28 and [the script] (tools/lemma28_radial.py, removed on 4 Nov 2026 by the rule that checking scripts are not kept; editorial change of the link only). The three items marked Error concern statements in the file: the identification of the boundary channels with a singular vector, a mislabelled table column, and the scope of the Correction and Outcome paragraphs. The matrix of Proposition 28.13 and its closed form are confirmed.

Count: 3 errors; 3 correct with amendment; 0 correct without amendment. The count refers to the six numbered items.

1. Slice operator — Correct with amendment.

With Hermitian Clifford generators $c_j c_k + c_k c_j = 2\delta_{jk}$, the spin connection is

$$\nabla_i = E_i - \frac{1}{4}\Gamma_{ijk} c_j c_k, \qquad \Gamma_{ijk} = \langle \nabla_{E_i} E_j, E_k \rangle.$$

Indeed, the commutator of the connection matrix with c_j gives $\Gamma_{ijk} c_k$. The Koszul expression in [the script] (tools/lemma28_radial.py, removed on 4 Nov 2026 by the rule that checking scripts are not kept; editorial change of the link only) has the correct order of indices. The $\mathfrak{h}$-part of a bracket is omitted in these reductive-frame coefficients; the H-action remains necessary in the equivariance condition.

For $s_v(g) = F\lambda(g)^{-1}v$, differentiation along $g\exp(te_i)$ gives $-FV_i$, with the appropriate scale divisor. The pull-back connection contributes $+E_i F/\mathrm{scale}_i$ in the vertical directions. Its sign in the script is consistent with the spin term.

The displayed radial form is correct for the script's $h = 1$ metric after absorbing the half-density. For the general metric in Theorem 28.9, let

$$u = h^2 \rho \, \psi.$$

Before this conjugation the radial derivative is

$$iZ \left(\partial_x + 2\frac{h'}{h} + \frac{\rho'}{\rho} \right).$$

After it, there is no extra h'/h or ρ'/ρ term, and the precise operator is

$$T = iZ\partial_x + \frac{A_0}{h(x)} + \frac{A_1}{\rho(x)} + \frac{\rho(x)}{h(x)^2} A_2 + i\phi(x)\tau.$$

A collar modification of the connection adds $-s(x)A_E/\rho(x)$. Thus the notation $D_6(\rho(x))$ in Result 28.5 needs the $h = 1$ qualification. The file should also distinguish its Hermitian Clifford convention from the skew-Clifford $c(dr)$ used in Theorem 28.9.

2. Exact tests — Correct with amendment.

The stated anticommutation and Casimir identities have the right form:

$$\{A_0, A_1\} = 0, \qquad D^2_{S^4, \tau_{\mathrm{tw}}} = \mathrm{Cas}_G(V) + \frac{3}{2} - \mathrm{Cas}_K(\tau_{\mathrm{tw}}).$$

Here τ_{tw} denotes the relevant K-twist; it should be distinguished from the Higgs matrix τ. Python reproduced the printed A_0^2 spectra and anticommutator norms in all four representations tested by the script. The zero in the A_0^2 spectrum on the **20** by itself is not a core zero mode of the full normal problem.

For the Fubini–Study metric with base S^4 of radius 1, the fibre radius is $\rho = 1$. The homogeneous strict nearly Kähler metric has $\rho = 1/\sqrt{2}$. More generally these radii are h and $h/\sqrt{2}$. The squashing conventions and the two special metrics are described in Appendix A.1, especially (A.1).

At $\rho = 1$, $j(j+4)$, $j = 0, 1, \ldots$, is the correct D^2 eigenvalue on $(0,0)$-forms with values in $\mathcal{O}(1)$. It is not the spectrum of the *entire* $\mathcal{O}(3)$-twisted spinor bundle: the **20** also contains $D^2 = 10$. The last sentence of the Correction paragraph needs that restriction. In addition, the script prints the Casimir and Fubini–Study spectra; it does not assert their expected numerical values.

The tests do not separately determine all three coefficients. For example, as an algebraic test of the listed identities, replacing

$$A_1 \mapsto tA_1, \qquad A_2 \mapsto A_2 + (1-t)A_1$$

preserves $\{A_0, A_1\} = 0$ and the tested line-connection operator at $\rho = 1$. The curvature construction supplies an additional test of A_2. With $a, b = 0, \ldots, 3$ and $\alpha = 4, 5$, Koszul's formula gives

$$A_2 = -\frac{i}{8} \sum_{a,b,\alpha} C^{\alpha}_{ab} c_a c_b c_\alpha.$$

The nonzero C^{α}_{ab} for $a < b$ are $C^5_{02} = -1$, $C^4_{03} = -1$, $C^4_{12} = 1$, and $C^5_{13} = -1$. An in-memory Python calculation compared this contraction on the full eight-dimensional spinor space with the script's ρ-coefficient. The Frobenius residual was 4.8×10^{-16}; reversing the coefficient gave residual 2.83. This check shares the script's Lie-bracket input, while using a separate curvature contraction. It is a numerical check of the displayed algebraic derivation.

3. Proposition 28.13 — Error.

The finite wall matrix, its kernel, and its trace calculation are correct for the channel convention used in that calculation. The error is its identification of those channels with the right singular vector for the boundary condition stated in Theorem 28.9.

For the stated $\hat{\Gamma} = \mathrm{diag}(1, -1)$ and $\hat{\Gamma} Z = -1$, the boundary domains are

$$Zu = -u \quad \text{for } T, \qquad Zv = +v \quad \text{for } T^*.$$

Let $T_\eta = iZ\partial_x + D_6 + i\eta\phi\tau$, with $\phi > 0$, and write a wall profile as $e^{q(x-\ell)}$, $q > 0$. The leading equations give

$$q = -\eta\phi Z\tau \quad \text{for } T_\eta, \qquad q = +\eta\phi Z\tau \quad \text{for } T_\eta^*.$$

Consequently, with the **positive mass and boundary condition of Theorem 28.9**, the right wall channels have $Z = -1, \tau > 0$, and their fibre trace is $+w$ in the script's $\hat{t}$ convention. The $Z = +1, \tau < 0$ channels of Proposition 28.13 give $-w$; they describe the adjoint wall problem for the reversed mass, or a problem with the opposite wall condition. Python reproduced $+2/3$ and $-2/3$, respectively, at $\rho = 1$. Changing the mass sign alone cannot make a right T-domain vector satisfy $Z = +1$.

For the matrix calculation, let $E_{s,t;j}$ be the rank-one map from input basis vector j to spinor basis vector s tensored with twist basis vector t, using the zero-based bases of the script. Choose

$$(f_0, f_1, f_2) = (E_{2,2;2}, E_{2,3;3}, E_{4,2;3}),$$
$$(g_0, g_1) = \left(E_{3,2;3}, \frac{E_{0,2;0} + E_{6,2;1}}{\sqrt{2}} \right).$$

Direct Clifford contraction, with the stable-profile overlaps, gives

$$\begin{pmatrix} 1/\rho & -3/(2\rho) & -(\rho + 1/\rho) \\ -i\sqrt{2} & 0 & -i\sqrt{2} \end{pmatrix}.$$

Reordering and changing unit phases gives the displayed $A(\rho)$. The entry $3/(2\rho)$, for example, contains the $\sqrt{3}/\rho$ vertical connection matrix element and the $\sqrt{3}/2$ overlap. Python/SymPy reproduced this matrix using exact symbolic arithmetic from the bracket and Clifford data.

Its kernel is as stated. The $-3/2$ channel and one $-1/2$ channel have input support $(1)_{-1}$, while the remaining channel has support $(1)_{+1}$; their output spaces are orthogonal. This yields

$$w(\rho) = \frac{2(\rho^2 + 2)^2}{9 + 2(\rho^2 + 2)^2},$$

including $w(0) = 8/17$, $w(1) = 2/3$, and $w(\infty) = 1$. For horizontal radius h, replace ρ^2 in this formula by ρ^2/h^2.

The line-preserving connection replaces the first entry of the first row by zero. Its remaining block has the stated sum and product of squared singular values, is invertible for $\rho > 0$, and has squared singular values $3, 6$ at $\rho = 1$. The Fubini–Study cohomology statement in the proposition has the stated scope.

For fixed positive radius, the overlap matrix is the correct **leading formal wall operator**. Since its kernel on the **4** is one-dimensional and separated from the other two singular values, the next correction is $O(M^{-1})$ in the fixed-block expansion and does not change its limiting kernel. A uniform global reduction still needs proof, as the file acknowledges.

The sentence calling 1.863 and 2.789 "the next values on the 4" needs to say **wall values**. In a Python radial calculation at $\phi_0 = 40$, $\ell = 3$, the levels were approximately 1.865 at the wall, 2.000 at the core, and 2.789 at the wall.

4. Result 28.5 — Error.

The Python rerun reproduced the main tables. The origin exponents are

$$\pm 1, \quad \pm 2 \text{ with multiplicity three}, \quad \pm 3.$$

For $\phi_0 = 3, 4, 5$, the measured decay rates in the first table agree with the corresponding constant-coefficient values $0.8557, 1.1514, 1.4762$ to the stated precision. At fixed ρ, the slow eigenvalues have the reported expansion

$$\kappa = \frac{\phi}{2} - \frac{2}{\rho} + O(\phi^{-1}).$$

For example, Python gave 30.0663 at $(\phi, \rho) = (64, 1)$ and 31.0625 at $(64, 2)$. The exponent comparison is numerical evidence for the displayed profiles; it is not a general tunnelling theorem.

The table has a numerical mesh floor, but its size depends on parameters and the mesh. At $(\phi_0, \ell) = (6, 4)$, Python gave lowest singular values $0.00548, 0.00401, 0.00355$ for $N = 2500, 5000, 10000$. These values do not establish a zero continuum value. Also, the script uses three-point quadrature for non-polynomial coefficients. Rayleigh quotients would give rigorous upper bounds with the exact quadratic form; the quadrature results are numerical approximations to those bounds.

The substantive table error is the heading **"sup of the density ratio."** The script computes

$$\frac{\max_x |\operatorname{tr}(B(x)\hat{t})|}{\max_x \operatorname{tr} B(x)},$$

not the supremum of the pointwise ratio. At $\ell = 3$, Python gave:

ϕ_0	Ratio of maxima reported	Pointwise maximum of $\lvert\operatorname{tr}(B\hat{t})\rvert / \operatorname{tr} B$
4	0.00648	0.849
5	0.00197	0.856
6	0.000473	0.858

The pointwise maximum occurs at the outer boundary, where the core density is small. The integrated density values -0.005707, -0.001836, and -0.000455 reproduce the table. Their ratios to λ^2 are empirical observations for these parameters, not a proved $O(\lambda^2)$ asymptotic. They support the hypothesis used by Proposition 28.16 for the numerical modes; numerical nonzero values alone do not prove the exact finite-mass algebra. At $\phi_0 = 16, 40$, the reported integrated values are around 10^{-11}, so their signs in these calculations do not establish nonvanishing.

For the other representations, the zero multiplicities on $\mathbf{1}, \mathbf{5}, \mathbf{10}, \mathbf{14}, \mathbf{35}$ follow from the central character. Python reproduced the reported wall/core identifications on $\mathbf{16}, \mathbf{20}, \mathbf{40}$, including core values close to $\sqrt{5}, \sqrt{8}, \sqrt{12}$. For **64**, an additional in-memory Python construction gave a small-radius wall value approaching $\sqrt{15}$. A scan of **401 radii** in $[0.1, 20]$ found no zero on any of the four representations; this supports a sampled statement and does not prove nonvanishing over the whole interval.

Two profile qualifications matter. The reported $\rho(x) = \tanh x$ metric is not exactly a product near the finite wall. The displayed tanh collar switch is nonzero, though tiny, at $x = 0$; an actual smooth extension with the pull-back connection near the core needs a cutoff that vanishes on a neighbourhood of zero. The rounded collar values therefore support the limiting model and do not establish exact finite-mass equality.

5. Correction, Outcome, and Conjecture — Error.

In the Correction paragraph, the old sign error and the withdrawal of the affected numerical claims are supported. The statement that **every number** was wrong is too broad: the previous Result 28.5 already listed the same origin exponents, and some displayed values agree after rounding. The Astra report says that its referee did not reproduce the old boundary-value numbers and that its proofs did not use them; this review did not repeat those proofs. The final sentence also needs the $(0,0)$-form restriction described in item 2.

For the four Outcome items:

1. The compression-map mechanism has the stated scope for the **prescribed** windows of Theorem 28.21.
2. The four core states are established for the normal model and its induced core operator. The displayed boundary weight is a finite wall-model result; convergence of an actual spectral mode remains conditional on the missing analysis. The boundary-sign and h-scale amendments above also apply.
3. The isolated window remains a Conjecture. Besides coercivity, the normal reductions, quasimode errors, and matching must be controlled.
4. The asserted exact finite-mass algebra is stronger than the numerical evidence. The table supports quaternionic-symmetry breaking at the resolved sample points, conditional on the continuum nonvanishing required by Proposition 28.16. It does not establish the exact algebra at every mass reported.

Conjecture 28.6 is compatible with Counterexample 28.17 when it asserts convergence of **compression maps**: convergence of those maps does not imply convergence of their generated algebras. The conjecture needs the boundary-sign and horizontal-scale conventions above. Its exponential statement in ℓ also needs a specified family of metrics and mass profiles with a uniform positive decay rate; Theorem 28.9 permits more general profiles.

6. Concentration literature — Correct with amendment.

Let $\mathcal{A} = i\Phi$ and let $\sigma(\xi)$ be the symbol of the self-adjoint Dirac operator. Since Φ acts on the twist factor, it commutes with the Clifford symbol, and

$$\mathcal{A}^*\sigma(\xi) = \sigma(\xi)^*\mathcal{A}.$$

Thus this is a concentrating pair in the sense of Maridakis, arXiv:1510.07004v1, Definition 2.1 and Lemma 2.2. The commutator $[D,\Phi]$ is order zero, and the doubled self-adjoint operator satisfies the corresponding anticommutation condition. On a closed manifold, the concentration principle applies to suitable low modes near the singular set of Φ.

The stronger spectral-separation theorem of Maridakis, arXiv:2110.11654v2, §2, Assumption 3 and the Spectral Separation Theorem has an additional hypothesis that fails for this spin-3/2 mass. At the core, its required direction-independent leading endomorphism would have to represent $(n\cdot J)^2$ for every normal unit vector n. These matrices depend on n: in a $J_3 = 3/2$ state, the expectations of J_3^2 and J_1^2 are $9/4$ and $3/4$. An anisotropic extension of that theorem, or a separate core analysis, is therefore needed. Prokhorenkov–Richardson, arXiv:math/0307251v3, §§2–3 supplies the related symbol condition and treats its stated localization setting.

Adding the wall requires boundary estimates and the stable boundary model. The closed-manifold energy argument acquires a boundary term: on a half-line,

$$\|(\partial_t + M)f\|^2 = \|f'\|^2 + M^2\|f\|^2 - M|f(0)|^2,$$

and e^{-Mt} is a boundary-localized zero mode although the mass has no zero there. A proof for this problem must combine the core analysis with that boundary layer, control the $O(M^{-1})$ reduction errors, and establish coercivity on the complement uniformly across representations.

For the source order, the arXiv LaTeX-source endpoints were requested first, but the available reader could not open their compressed responses and direct network retrieval failed. We then read the cited sections

of the three arXiv HTML versions identified above. We did **not** read the LaTeX sources or PDFs of those three papers. The separate Fubini–Study-radius citation above was checked in its PDF appendix.

Part R16 (Terra referee, lemma 29)

Referee report on lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_29.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. Item 10: correct, the definition of A_eff as a chiral map made K(rho) impossible as written, while the displayed operator was the intended one; corrected. Sign of the fibre part: correct, -2/3 from the kernel vector (-2, 1, 1), with channel weights 4, 1, 1 and fibre values -1, -1, +1; the squared values (45 +- 3 sqrt 33)/8 were recomputed by hand. Items 2, 4 to 9, 11, 12: applied. Item 9 also supplies the density B_w, checked at w = 1 and against the trace. Item 13: the identifiers were checked on arXiv. A referee that ran no computation and reports no counterexample has found no error in case (b); that is weaker than a proof by a second author, and the corrected case (a) has not been read again.

The principal error is in case (a): the displayed definition of A_{eff} makes Condition $K(1)$ impossible as written. There is also a sign error in the reported fibre part: at $\rho = 1$ it is $-2/3$. Both have identifiable corrections; neither supplies a counterexample to the proposed eight-state count.

The case-(b) coercivity argument survives the main cutoff and projection checks, subject to the domain, gauge and constant clarifications below. No Python calculation was run. The table was compared with the stated limiting values, but its computations were not independently reproduced.

1. **CONFIRMED — Proposition 29.1.** For the connection metric with $h = 1$ and $\rho = r$, the normal fibres are flat and totally geodesic. In a horizontal/vertical orthonormal frame, the difference between the Levi-Civita connection and the split connection is the O'Neill tensor. Its curvature component is linear in y, and its spin contraction gives, in the convention of Proposition 29.1,

$$R = -\frac{i}{8} \sum_{a,b,\alpha} \langle [\tilde{e}_a, \tilde{e}_b], f_\alpha \rangle c(e_a) c(e_b) c(f_\alpha).$$

   Thus $T_M = T_0 + R$ exactly on the tube. There is no additional first-order term for this metric and the pull-back connection. In the moving radial frame this is rA_2; the apparent polar mean-curvature term belongs to the ordinary vertical Dirac operator.

   The bracket normalization used in the radial system reduces A_2, on the nontrivial horizontal chirality, to a unitary conjugate of

$$\tfrac{1}{2}(s_1 \otimes \sigma_1 + s_2 \otimes \sigma_2),$$

   with s_i Pauli matrices. Its norm is 1; it vanishes on the other horizontal chirality. Hence $\|R\| \le |y|$.

   The common principal connection causes no extra derivative of $\widehat{Q}$. In the equivariant-map realization, the horizontal fields differentiate the G argument and $\widehat{Q}$ acts by the same operator on the fibre at every g. Diagonal K-invariance of the actions on y, the normal spinors and E makes this construction descend. Individual horizontal derivatives have a frame index that transforms under K; the contracted D_H is equivariant.

   With $c(f_\alpha) = \gamma_5 \sigma_\alpha$, the relevant algebra is

$$D_H \widehat{Q} = \widehat{Q}^* D_H, \qquad D_H \widehat{Q}^* = \widehat{Q} D_H.$$

It gives both displayed sums of squares and commutation of D_H^2 with the two normal squares. Finally,

$$\Sigma_7|_K = ((2,1)\oplus(1,2))\otimes(1,2) = (2,2)\oplus(1,3)\oplus(1,1),$$

so the stated spin representation is correct.

2. **AMENDMENT — Proposition 29.2.** The spectrum and kernel conclusions follow, with a distinction between two projections. The projection P_W onto the normal K-representation $W_0 = (1,3)\oplus(1,1)$, extended over *both* base chiralities, commutes with D_H. The projection onto the actual fibre kernel of $\widehat{Q}$ also selects the chirality $\gamma = -\epsilon$:

$$P_0 = P_{\gamma=-\epsilon}\otimes P_W.$$

This projection need not commute with D_H. Rather, $D_H P_0 = P_1 D_H$, where P_1 projects onto the fibre kernel of $\widehat{Q}^*$, and P_0 commutes with D_H^2. The proof should use P_0 and the latter commutation for the normal spectral gap.

For the stated normalization, the twisted round-S^4 formula is

$$D_H^2 = C_2^G(V) + \tfrac{3}{2} - C_2^K(j), \qquad C_2^K(j) = 2j(j+1).$$

The twist has $j = 0, 1$. Only representations with odd central character occur. The least G-Casimir is $5/2$ on the 4; the next is $15/2$ on the 16. The $j = 1$ zero on the 4 occurs in the base chirality of type $(1,2)$, exactly the chirality selected by (H4). The adjoint normal kernel selects the opposite base chirality, where that zero is absent. The $j = 0$ part on the 4 gives the next squared value 4.

On noncompact N, the calculation should first be stated on compactly supported smooth sections and extended through the closed graph forms. The doubled model is essentially self-adjoint on that core. The normal estimate $(y\cdot J)^2 \geq |y|^2/4$ gives compact normal resolvent; the compact base and the commuting squared decomposition give the required discrete model spectrum. No boundary condition is imposed at infinity.

3. **CONFIRMED — Lemma 29.3.** Coupling spin 1/2 to spin 3/2 gives

$$\sigma\cdot J = F^2 - \tfrac{3}{4} - \tfrac{15}{4},$$

so its eigenvalues are $3/2$ and $-5/2$. For each y, rotational invariance gives eigenvalues $\pm|y|/2, \pm 3|y|/2$ for $y\cdot J$; hence the squared lower bound is pointwise valid. Consequently

$$\frac{M^2}{4}\||y|u\|^2 \leq \|T_0 u\|^2 + \frac{5M}{2}\|u\|^2,$$

which gives exactly the stated constants after taking square roots. The adjoint estimate follows in the same way. The positive μ_1, rather than its numerical value, is all that the subsequent proof needs.

4. **AMENDMENT — Lemma 29.4 and IMS.** Define C_Φ explicitly as a bound on the *Clifford contraction*

$$\left\| \sum_i c(e_i)\nabla_{e_i}(f\, n\cdot J) \right\|_{\mathrm{op}}.$$

This removes the ambiguity in writing only $|\nabla(f\, n\cdot J)|$. With that convention the bulk estimate follows from expansion of $(D+i\Phi)^*(D+i\Phi)$.

The IMS step is valid. The derivatives of all three cutoffs are supported in a compact annulus away from both ends; the bulk estimate controls their error by $CM^{-2}(\|T_M u\|^2 + \|u\|^2)$. Dropping the nonnegative $\|T_M\chi_b u\|^2$ is allowed because the same bulk estimate separately controls $\|\chi_b u\|$. Scalar radial cutoffs preserve both wall domains.

The comparison with the model is also correct: extension by zero puts $\chi_c u$ in the model domain, and

$$u \perp \chi_c \ker T_0 \quad\Longrightarrow\quad \Pi(\chi_c u) = 0.$$

The analogous wall orthogonality is exact.

5. **AMENDMENT — Lemma 29.5.** Parthasarathy's formula gives the displayed result. In the standard $SU(4)$ normalization, the central weight of the $\mathcal{O}(k)$ twist is represented by

$$\tfrac{k}{4}(1,1,1,-3),$$

whose squared length contributes $3k^2/8$ to the Casimir. The scalar curvature of the stated Fubini–Study metric is 12, giving $R/8 = 3/2$. Thus

$$D_k^2 = C_2(L) + \tfrac{3}{2} - \tfrac{3}{8}k^2.$$

The invariant unitary connection on each line is the Chern connection: the difference of two invariant line connections is an invariant one-form, and the H-isotropy representation has no trivial summand.

The eligibility argument works. Each of the two $SU(4)$ representations of dimension 20 below the relevant Casimir threshold restricts to $Sp(2)$ as $4 \oplus 16$. The $\mathcal{O}(3)$ slice contains the $Sp(2)$ representation 4 only once, and that occurrence is already the cohomological kernel; neither 20 can therefore occur. In the charge convention of the earlier report, $\mathcal{O}(3)$ has charge -3. The slice weights give its 4 once in $Z = +1$ and zero times in $Z = -1$, proving the kernel chirality without a numerical mode-position argument.

The correction is to "$\kappa_w = \sqrt{3}$, for all V": $\sqrt{3}$ is the **global uniform gap**, attained on the 4. It is not the lowest nonzero value on every isotypic component; the stated lowest wall value on the 16, for example, is $\sqrt{5}$.

6. **AMENDMENT — Lemma 29.6.** In radial parallel gauge the sum of squares and Robin analysis are correct. With $q = -\tau$, the free Robin zero channel is

$$(Z = -1, q > 0) \quad \text{for } T_M, \qquad (Z = +1, q > 0) \quad \text{for } T_M^*.$$

The first-order form $d - m|q|$ has the normalized zero profile $\sqrt{2m|q|}\, e^{m|q|(x-l)}$, and its orthogonal complement has spectrum starting at m^2q^2. The boundary condition kills the cross-term boundary pairing.

The collar gauge should be stated. For example, add $itq\,dr$, with a cutoff supported in the collar, to a line-preserving connection. This is invariant, unitary and satisfies (H3). In the original radial trivialization it changes the Dirac operator by $-tZq$; on the $q = 3/2$, $Z = +1$ line,

$$T^*_{M,\mathrm{new}}(pg) = -\tfrac{3}{2}t\,pg.$$

Radial parallel transport removes this term and restores the displayed profile. Thus the geometric estimate persists, while the displayed formula and trial state require that gauge or the corresponding parallel-transport factor.

The charge calculation in item 5 establishes the relative core/wall sign. The index discussion in Remark 29.12 gives a consistency check; it does not by itself identify the local chirality. If the $\mathcal{O}(3)$ zero mode instead lay in $Z = -1$ with $q > 0$, pg would be a wall zero mode for T_M, contradicting the positive lower bound in part 1 of Lemma 29.6.

7. **AMENDMENT — Theorem 29.7 and Lemma 28.14.** Lemma 28.14 is correct. If the trial range has dimension eight and $\|\widehat{D}U_M\| < \delta$, its spectral projection onto $(-\delta, \delta)$ has dimension at least eight. If that projection had dimension greater than eight, it would contain a nonzero vector perpendicular to the trial range, contradicting the complementary bound. The spectral theorem then gives the stated angle estimate.

With the projection and collar-gauge clarifications above, the case-(b) IMS argument supplies the hypotheses for each fixed $0 < \delta < \sqrt{3}$. The core trial defect is $O(M^{-1/2})$: Lemma 29.3 gives the required first moment even without invoking Gaussian decay. The wall defect is exponentially small in radial parallel gauge. No counterexample to the case-(b) count was found.

8. **AMENDMENT — Proposition 29.8.** The tunnelling identity is correct for a singular pair with real nonnegative λ. For a real scalar weight preserving the domain, the quadratic-form identity

$$\|T_M(\xi u)\|^2 = \lambda^2\|\xi u\|^2 + \|c(d\xi)u\|^2$$

is also correct.

The Agmon proof needs two explicit choices. First, choose the IMS cutoffs so that the region where the exponential weight varies is covered by the bulk estimate, including the transition near the collar. Second, the final cutoff loses the integral across its support. Apply the weight with a smaller parameter, for example $\eta' = \eta/2$, and choose the support width s so that

$$(1-\eta')\left(I - \int_{r_*}^{r_*+s} f\,dr\right) \geq (1-\eta)I, \qquad I = \int_{\varepsilon}^{l-a} f\,dr.$$

This supplies the exponent stated in the proposition. The absorption uses $\lambda \to 0$, supplied by the trial states.

The factor $1/2$ is justified uniformly across channels by $(n\cdot J)^2 \geq 1/4$; it does not require every channel to have eigenvalue $1/2$. For the subsequent $l \to \infty$ statement, require $M \geq M_\eta$ and uniform bounds on the endpoint geometry, cutoffs, connection and positive bulk mass. Translation invariance in the bulk alone should not stand in for those uniform hypotheses.

9. **AMENDMENT — Theorem 29.9.** The compression estimate follows from Theorem 28.15 after choosing equivariant unitary identifications of the varying window spaces with the model copies of 4, and after the applicable window theorem is established.

With the coordinate convention of lemma 28,

$$Q_B(a) = \int_{S^4} a(x)(1 - x\cdot\Gamma)\,d\mu(x), \qquad Q_B(x_a) = -\Gamma_a/5.$$

In case (b), $B(z) = 4p_z$, with $p_z \leq (1 - x\cdot\Gamma)/2$. This base-coordinate convention accounts for the sign when comparing with the coordinates in lemma 10.

More explicitly, if p_z is the coherent line selected by the $q = 3/2$ channel and p_{jz} its quaternionic partner, the case-(a) limiting density is

$$B_w(z) = 2((1+w)p_z + (1-w)p_{jz}).$$

It gives $Q_Y(x_a) = -\Gamma_a/5$ and the stated factor w on the y_T components. Since $w > 0$, the 5- and 10-components are present; together with independent endpoint restrictions they give the claimed real algebra. This is a limit-map statement.

10. **ERROR — Proposition 29.10 and Theorem 29.11 as written.** At the definition of A_{eff}, P contains only $Z = -1$ states and Y only $Z = +1$ states. Therefore

$$F = \Pi_Y(A_d + A_o)P : P \longrightarrow Y$$

is a chiral operator. It is not the self-adjoint operator subsequently described. If regarded as an operator on the ambient space, it kills all of Y; if regarded as $P \to Y$, its kernel lies in $Z = -1$. In either reading, the stated Condition $K(\rho)$, requiring its kernel to be exactly a 4 in $Z = +1$, cannot hold.

The correction is to define the full operator on $P \oplus Y$ by

$$\mathcal{A}_{\text{eff}} = \begin{pmatrix} 0 & F^* \\ F & 0 \end{pmatrix},$$

and impose $K(\rho)$ on this operator. The matrix $A(\rho)$ of Proposition 28.13 represents $F^* : Y \to P$ on the 4; F is its adjoint.

With this correction, the proposed block argument has the required structure. The profile overlap is $\sqrt{3}/2$. The stable-to-stable block has norm $\sqrt{3}/\rho$, so its contribution has norm $3/(2\rho)$; the full perturbation instead has $\|A_o\| = 2/\rho$, as required in the elimination estimate. The complementary block has inverse norm at most $(m/2 - \|A_o\|)^{-1}$ for large m, and its Schur correction is bounded by

$$\frac{\|A_o\|^2}{m/2 - \|A_o\|}.$$

On $V \neq 4$, the corrected effective operator at $\rho = 1$ is bounded below by $\sqrt{5} - 3/2$. On the 4, the nonzero squared singular values of $A(1)$ are

$$\frac{45 \pm 3\sqrt{33}}{8},$$

and its one-dimensional multiplicity kernel lies in $Z = +1$. This proves the intended corrected $K(1)$. The quoted 1.863 is the value on the 4; the displayed argument gives the smaller uniform lower bound $\sqrt{5} - 3/2$ off that representation. The cut-off trial state must use the exact corrected half-cylinder kernel; its finite 4-block equation gives exponential decay.

There is a separate sign error in the reported fibre part. At $\rho = 1$, the kernel vector is $(-2, 1, 1)$, and the signed fibre trace is

$$\frac{-4 - 1 + 1}{6} = -\frac{2}{3}.$$

In general it is $-w(\rho)$. The positive $w(\rho)$ is its magnitude.

11. **AMENDMENT — Compatibility of (H1)–(H4).** The hypotheses are compatible. A smooth positive ρ can equal r near zero and be constant on a disjoint collar; take $h = 1$. Likewise, $f = r$ near zero and $f = f_0 > 0$ on the collar gives $f(r) = r\,b(r^2)$ for a smooth positive b. The metric and mass extend smoothly across the zero section. Since X deformation retracts to S^4, its relevant $H^2(-; \mathbb{Z}_2)$ vanishes and the oriented tangent bundle is spin. Invariant Hermitian connections can be interpolated with a radial cutoff.

    The chirality convention should be explicit. If $\tau = n \cdot J$, then (H4) gives $q = -\gamma_b \tau$; the displayed convention $q = -\tau$ uses $\gamma_b = +1$. The collar expression also uses the radial gauge described in item 6.

12. **ERROR — Outcome and scope.** Outcome item 2 and Theorem 29.11 give the signed fibre part as $+2/3$; it is $-2/3$. The assertion that $K(1)$ holds also requires the corrected effective-operator definition in item 10. The nearest supported statement is the intended case-(a) result after that correction and the accompanying Schur-complement argument.

    Remark 29.12 should say that the opposite-sign **end-model modes** lie in the indicated kernels. At finite mass the full operator can have a nonzero tunnelling singular value, so those localized states need not lie in the exact kernels of T_M or T_M^*.

    The scope paragraph otherwise identifies the restrictions on metrics, the connector and finite-mass algebras. "Uniformity over representations" follows here from the stated full-space estimates and ellipticity of the corrected fixed boundary operator; it should not be attributed to the fixed-manifold setting alone. Away from $\rho = 1$, finiteness of the possible kernel representations follows from ellipticity, while an explicit representation cutoff for a practical finite check remains unwritten.

    The displayed **Python** table is consistent, to its shown precision, with the stated limits: in particular 2.2361, 2.8284 and 3.4641 agree with $\sqrt{5}, \sqrt{8}, \sqrt{12}$, and the larger $M = 32$ wall discrepancies are of the stated mesh scale. The table was not rerun. The quoted $\mu_1 \approx 1.02$ is a Python approximation; the two supplied tools do not themselves reproduce that oscillator calculation.

13. **AMENDMENT — Prior work.** General forms of two key identities already appear in the literature:

    - Maridakis, arXiv:1510.07004v1, Lemma 2.2 gives the concentrating-pair square and its zeroth-order cross term, which includes the bulk identity used here. Prokhorenkov–Richardson,

arXiv:math/0307251v3, §§2–3 treats the earlier Witten-deformation localization framework for the stated isolated-singularity setting.

- Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18 gives a horizontal/vertical Dirac factorization with an explicit curvature contraction. Its stated proper-submersion and dimension hypotheses differ from this seven-dimensional noncompact model; it is a close structural ancestor of Proposition 29.1.
- Bismut–Lebeau, *Complex immersions and Quillen metrics*, §§8–9, DOI 10.1007/BF02699352, develops rescaled normal operators and block estimates for complex immersions. Kottke, arXiv:0909.5661v2, Theorem 11 gives the families Callias index in terms of the positive eigenbundle; it supplies an index, rather than this boundary spectral window.
- Maridakis, arXiv:2110.11654v2, §2 proves a spectral-separation theorem under additional assumptions. Its Assumption 3 requires a direction-independent leading endomorphism; the spin-3/2 mass here has direction-dependent $(n \cdot J)^2$. Prokhorenkov–Richardson, arXiv:math/0610160v2, §3 treats equivariant Witten localization and normal models.

The targeted search covered Callias operators on bundles, submersion factorization, Bismut–Lebeau localization, equivariant Witten deformation, boundary chirality and a 2026 recency query. The cited sections were opened. Taubes's 1996 paper, DOI 10.4310/jdg/1214459411, was identified through Maridakis's reference chain, but its original full text was inaccessible in this search. Citation chains and the 298-page Bismut–Lebeau paper were not exhaustively reviewed. The search found prior general identities; it does not establish whether the specific eight-state theorem has appeared elsewhere.

The corrections in order of severity are:

1. Replace the definition of A_{eff} and $K(\rho)$ by the full chiral block, then restate the case-(a) proof.
2. Correct the signed case-(a) fibre part to $-w(\rho)$, in particular $-2/3$ at $\rho = 1$.
3. Specify radial parallel gauge, the model domains and the distinction between P_W and the one-chirality fibre-kernel projection.
4. Complete the Agmon cutoff argument with an η-margin and state the uniform hypotheses for varying l.

Part R17 (Terra referee, lemma 30 and the joint-limit note)

Referee report on lemma 30, the note and the corrected case (a) of lemma 29, by Terra 6.1

Model gpt-6.1-terra, effort max, read-only sandbox, run on 4 November 2026 with the prompt terra_check_30_note.md. The report below is verbatim, except that local absolute links were rewritten as relative links. No Python was run by the referee.

Assessment by Claude Opus 5.5, reading it against the text. All items applied, with these checks. Item 7: correct; the bulk interpolation allowed by (H3) need not keep j_E parallel. The fix was checked by hand: j_E (n . J) j_E^-1 = -n . J gives j_E P_q j_E^-1 = P_{-q}, so j_E is parallel for the line-preserving connection and for real affine interpolation with the pull-back connection, which is the interpolation in tools/lemma28_radial.py, and C then commutes with i Phi_M as well as with D. The unconditional bound eta_t = O(M^{-1/2}) is adopted, with its proof written out. Item 4: correct; lemma 24 has V = C^4 with gradings of type (2, 2). Item 9, first point: the counterexample to the implication from the hierarchy alone is correct. Item 5: the sign in Proposition 29.10 and the missing projection estimate are applied; the Weyl constant has the rank 64 = 8 x 4 x 2. Item 1: the gap is an equality, since the next level is (pi - theta_max)/l. Item 13: the identifiers not already in the record were checked on arXiv. The source-level attributions to Parthasarathy, Bismut-Lebeau and Kirchberg stay unread, as the referee says. This was the last Codex job available in the weekly quota, so these corrections have had no further independent reading.

The main mathematical gap is in Proposition 30.7: the allowed case-(b) connections need not preserve the

charge conjugation used in the proof. With an added compatibility hypothesis, the weighted argument gives the displayed exponential bound. A fresh reading of the corrected case (a) also found a remaining sign error in Proposition 29.10.

The interval and disc-bundle counting arguments otherwise support the stated joint limits, subject to the amendments below. No Python calculation was run in this review.

Part A: lemma 30

1. **AMENDMENT — Theorem 30.1.** The compression constant 2/3, the gap bound and both counts agree with Theorems 22.1 and 22.4. Add $l_* > 0$, $E_* > 0$, and the algebra and representation assumptions needed to define $Q_\varepsilon(a)$ and $\pi(a)$. Alternatively, state the compression estimate for a bounded even operator a, using its operator norm.

   There are $r/2$ singular values counted with multiplicity, and each gives two signed arithmetic progressions. These remain r **progressions with multiplicity** when singular values coincide or vanish. In particular, a zero singular value contributes two zero modes. Thus the moving-window count is r, and the fixed-energy leading term is

   $$\frac{2rl_*E}{\pi}\varepsilon^{-\alpha}.$$

   For small ε, the displayed lower bound for the first discarded level is in fact an equality:

   $$\Delta_\varepsilon = \frac{\pi}{l_\varepsilon} - \varepsilon\|D_*\|.$$

2. **AMENDMENT — Corollary 30.2.** The two endpoint pairs are correct for the **universal fully massive family**, with both members of every Standard Model doublet assigned the same Dirac mass and no Majorana mass. In that family the two commutator quotients are $2m^2$ and $16m^2/5$; multiplying them by $(3 - d_{\text{eff}})/2$ gives $(3, 24/5)m^2$ as $d_{\text{eff}} \to 0$ and $(2, 16/5)m^2$ as $d_{\text{eff}} \to 1$.

   Lemma 25 concerns the leading **Taylor quotient** for the heat-cutoff spectral action, with an even vector generator constant in the transported frame and an enlarged admissible gauge algebra. The interval poles belong to the separately specified five-dimensional Yang–Mills problem. The coefficient comparison holds at leading order; it does not identify the full spectral-action pole problem with that Yang–Mills problem.

   The path is correct:

   $$s = l_\varepsilon E_\varepsilon = l_* E_* \varepsilon^{\beta-\alpha} \longrightarrow 0, \qquad \frac{m_\varepsilon}{E_\varepsilon} \longrightarrow 0.$$

   At fixed $\Lambda > 0$, $s \to \infty$ and $m_\varepsilon/\Lambda \to 0$. Lemma 25 states its expansion first at fixed s; a joint-limit use should add a uniform remainder argument, or state explicitly that the displayed endpoints concern the leading coefficient. Varying the action cutoff is an additional prescription, as the corollary says.

3. **AMENDMENT — Proposition 30.3.** The root and asymptotics follow from Theorem 9.4 and Lemma 9.5. For $c > 1$, the first oscillatory root satisfies

   $$\pi < k_1 l < \frac{3\pi}{2}, \qquad \frac{\Delta}{\mu} = \sqrt{1 + \left(\frac{k_1 l}{c}\right)^2} \longrightarrow 1.$$

   The two tail identities follow from $\mu = \kappa \coth s$ and $m = \kappa/\sinh s$.

   The localized family needs a **new length**, since its length cannot also be the $l_*\varepsilon^{-\alpha}$ of Theorem 30.1. State

   $$l_\varepsilon^{\text{loc}} = \frac{(1-\alpha)\log(1/\varepsilon) + c_0}{\mu_*\varepsilon^\alpha},$$

   with $\mu_* > 0$ and $0 < \alpha < 1$. The reuse of l_ε is also misleading in the note.

4. **ERROR — Corollary 30.4 for arbitrary finite data.** The cylinder in lemma 24 has $\ker A \simeq \mathbb{C}^4$, with wall gradings having two positive and two negative directions. It does not realize an arbitrary dimension r from Theorem 30.1. For example, finite data with $H_+ = H_- = \mathbb{C}^3$ and $D_* = 0$ satisfy the interval hypotheses and have $r = 6$; the stated cylinder has a four-dimensional tangential kernel.

   Restrict the corollary to the compatible $r = 4$ data, or construct and state an enlarged tangential bundle for other dimensions. The gap estimate then follows from lemma 24. The seven-dimensional count needs an additional Weyl argument: Dirichlet and Neumann bracketing of the squared product operator give the same leading term as $l_\varepsilon, r_\varepsilon \to \infty$, proportional to $l_\varepsilon r_\varepsilon^6 E^7$. Lemma 24's gap estimate alone does not prove that asymptotic.

5. **AMENDMENT — Theorem 30.5 and corrected case (a).** For a fixed shape, it is legitimate to substitute $M = M_\varepsilon \to \infty$ into Theorem 29.7 or 29.11. Proposition 29.8 supplies M_η, C_η independent of M after the shape and η are fixed. Substitution gives $\lambda_M / R_\varepsilon \le C_\eta \varepsilon$. No uniformity in a varying shape is needed for this family.

   The domain of $\widehat{D}_M$ is independent of M: the wall projector is fixed and MV is bounded, with $\|V\| \le (3/2) \sup f$. Min–max applied to the singular values gives the displayed comparison of the counting functions of $|\widehat{D}_M|$ and $|\widehat{D}_0|$. The local self-adjoint elliptic boundary problem has the usual leading Weyl law. Counting both signs and complex multiplicity,

$$C_X = \frac{64\,\omega_7\ \mathrm{Vol}_g(X)}{(2\pi)^7},$$

   since the seven-dimensional spinor bundle has complex rank 8, E rank 4, and the operator is doubled. The estimate $M_\varepsilon / R_\varepsilon \to 0$ makes the bounded perturbation negligible at the counting threshold. The physical first discarded level also tends to zero: its shape-unit value is at most $s_9(\widehat{D}_0) + cM_\varepsilon$.

   Dilation is exact under the canonical spinor identification and the L^2 unitary factor $R^{-7/2}$. With physical radial coordinate $s = Rr$, the potential is

$$\frac{M}{R}\, f(s/R)\, n \cdot J.$$

   A radial half-density normalized with respect to ds acquires the corresponding $R^{-1/2}$ factor. The compression estimate uses C^1 norms in fixed shape coordinates. Add $0 < \delta$, $E_* > 0$, and state that if $m_\varepsilon = 0$, zero has total multiplicity eight.

   Fresh check of case (a). The corrected chiral definition of A_{eff}, the profile overlap $\sqrt{3}/2$, the perturbation bound $3/2$, and the squared nonzero singular values

$$\frac{45 \pm 3\sqrt{33}}{8}$$

   are consistent. The displayed matrix at $\rho = 1$ has kernel $(-2, 1, 1)$, giving signed fibre part $(-4 - 1 + 1)/6 = -2/3$. However, Proposition 29.10 still says that the fibre part tends to $+w(\rho)$. It must say $-w(\rho)$. Theorem 29.11 has the correct sign.

   The adjoint estimate in Proposition 29.10 also needs one explicit step. Orthogonality to the *exact* half-cylinder kernel does not immediately give orthogonality to $\ker F^*$. The Schur-complement graph estimate gives $\|P_{\ker T_M^*} - P_{\ker F^*}\| = O(1/m)$; inserting this projection estimate supplies the required $O(1/m)$ loss. With that addition, no further error was found in the corrected case-(a) block argument.

6. **CONFIRMED — Proposition 30.6.** Normalized angular Haar measure and Schur's lemma give

$$Q(f) = \frac{4}{\int \operatorname{tr} B} \int\!\!\int f(r, g) \lambda(g)^{-1} B(r) \lambda(g)\, dg\, dr.$$

   Quaternionic conjugation exchanges b_+ and b_-, so $\|B - P_{\mathbb{H}} B\| = |b_+ - b_-|/2$. This gives $2\eta_t$. Each compression of a real f with $\|f\|_\infty \le 1$ is a contraction; telescoping against its j-conjugate gives $2k\eta_t$ for a product of k such compressions.

7. **ERROR — Proposition 30.7 as stated for all allowed connections.** The proof assumes $CT_M = T_M C$ throughout the interior. Setting 29.0 permits any invariant unitary interpolation in the bulk; it does not require that interpolation to preserve j_E.

   This is a failure of a stated premise in the proof. For an allowed connection that is pull-back on a bulk annulus, put $q = n \cdot J$ there and add the invariant skew-Hermitian connection form

   $$i\, h(r)\, \nabla^0(q^2),$$

   with real h supported inside the annulus. It has no radial component and leaves both end hypotheses intact. Since $j_E q^2 j_E^{-1} = q^2$, antiunitarity sends this added form to its negative. Consequently j_E is not parallel and C need not commute with the resulting twisted Dirac operator. This example shows the failure of the asserted symmetry; it does not by itself establish a counterexample to the exponential estimate.

   Add $\nabla^E j_E = 0$, or another explicit hypothesis proving the same global symmetry. Under that hypothesis, the other checks work as follows.

   - $C^2 = j^2 = -1$, so $F \mapsto CFj^{-1}$ is an antilinear involution on $\mathrm{Hom}_G(\mathbf{4}, -)$. On the core it preserves D_H, the full fibre term $i\widehat{Q}$, and R. With the conventions of Proposition 29.1, it anticommutes with $\widehat{Q}$ itself, because it is antilinear.
   - The phase can make $\langle \chi_c F_0, u\rangle$ real, and then $\Pi(\chi_c \zeta) = 0$. The qualification by χ_c is essential.
   - On $r \le l - a$, $g + \tilde{g} = G$. The cut-off $\xi\zeta$ vanishes near the wall and belongs to the required domain. The bulk first-order term has a positive η-margin; the core estimate uses the projection condition. The derivative of the wall cut-off is supported where $g = G$, so its contribution is $O(e^{-G})$. Thus the proposed argument gives $\|\xi\zeta\| = O(e^{-G})$.
   - Pointwise,

     $$|b_+ - b_-| \le (\|F\| + \|CF\|)\|\zeta\|.$$

     Weighted Cauchy–Schwarz on $r \le l - a$, and the L^2 tail bound beyond $l - a$, both give $O(e^{-2G})$.

   Therefore, **conditional on global C-preservation**, the proof gives

   $$\eta_t \le C_\eta \exp\left[-(1-\eta) M \int_{2e}^{l-a} f\, dr\right].$$

   The integral starts at $2e$, whereas Proposition 29.8 displays the tunnelling bound starting at e; the stated exponent is therefore not literally twice that displayed tunnelling exponent. A smaller core cut-off can be chosen if the comparison is to use the same starting radius.

   Under the original general connection hypotheses, the window estimate still gives $\eta_t = O(M^{-1/2})$: choose an equivariant normalized core model fixed by the local real structure and use the L^2 distance to that model from Theorem 29.7. Finally, an exponential upper bound expressed in M does not imply $O(\lambda^2)$ without a lower bound on λ.

   The Python samples cited at lemma 30, line 106 vary the length at fixed mass and use profiles outside Setting 29.0. They support a qualitative comparison and do not test the stated $M \to \infty$ theorem.

8. **AMENDMENT — Remark 30.8.** The vector-sector and connector obligations are accurately identified. The gauge formula assumes a fixed microscopic coupling, a constant internal Yang–Mills profile and the stated normalization; then $g_4 \propto R^{-7/2}$.

   For the toy potential, require $a, b, p, q > 0$. Its unique minimum is

   $$r_\varepsilon = \left(\frac{ap}{bq}\right)^{1/(p+q)} \varepsilon^{-2/(p+q)},$$

   and $p + q > 2$ gives $0 < \alpha < 1$. The complete-manifold sentence just before the remark also needs qualification: global charge conjugation gives a real structure on the multiplicity space, but the provided argument does not establish that this space is one-dimensional for **every** M or for every completion. Exact quaternionicity of a selected core copy follows if that multiplicity is one.

Part B: the note

9. **ERROR — Several statements exceed their supplied hypotheses.** The following corrections are needed in joint_limit_note.tex.

   - **Hierarchy, lines 40–44.** The hierarchy alone does not imply that the fixed-energy count diverges. For a compact-resolvent counterexample, take signed spectrum

     $$0, \quad \pm\varepsilon^{\alpha}, \quad \pm 1, \quad \pm 2, \ldots$$

     and $E_\varepsilon = \varepsilon^\beta$, $\alpha < \beta < 1$. The hierarchy holds with one retained zero mode, while $N_\varepsilon(E) = 3$ for fixed $0 < E < 1$ and small ε. State divergence as a separate property proved for the specific families.
   - **Interval and cylinder, lines 50–79.** Add the representation assumptions for the compression bound, separate the localized length from the angle-family length, and restrict the unmodified cylinder to compatible four-dimensional kernel data, as in items 1, 3 and 4. State the scope of the heat-cutoff Taylor quotient from item 2.
   - **Disc-bundle hypotheses, lines 83–90.** Define $S = G \times_K (1, 2)$ and its pull-back in $E = S^3(S)$; state the round S^4 radius, smoothness and positivity of ρ and f, $f_0 > 0$, and $Sp(2)$-invariance of the connection. The displayed metric with $\rho = r$ does encode flat normal fibres near the core; the note also explicitly states constant f and the Fubini–Study radius on the collar. Case (a) in lemma 29 assumes the pull-back connection throughout, whereas the note only specifies it near the two ends. A bulk interpolation may be treatable by the bulk estimate, but that extension needs to be stated and proved.
   - **Sign and localization, lines 87–90.** The sign selects the core kernel of the **end model**. At finite M, the full T_M can have nonzero tunnelling singular value. Describe the two four-dimensional subspaces by their proved proximity to the cut-off core and wall models, rather than as exact states at the two ends.
   - **Core proof, line 99.** The gap of the first-order fibre operator is of order $\sqrt{M}$; the gap of its square is of order M. D_H exchanges the relevant base chiralities and need not preserve the actual one-chirality fibre kernel. The argument uses the normal-mode projection extended over both chiralities, or the commutation of the actual kernel projection with D_H^2.
   - **Bulk and wall, lines 101–110.** Replace the variable-f lower bound by an integral bound or a bound using the minimum of f on the support. Restrict the asserted wall gap to the relevant stable $q > 0$ channels: the full line sum also includes the $\mathcal{O}(-3)$ zero modes. The case-(a) value $-w(\rho)$ is the limiting effective-wall value, with a finite-M correction.
   - **Tunnelling and joint limit, lines 128–162.** Define the integration endpoints and the fixed cut-offs e, a, rather than writing only "over the bulk." State the equivariant identifications and the shape-coordinate C^1 norm in the compression estimate. Include the zero-mass multiplicity qualification from item 5.
   - **Defect, lines 166–175.** Add the global charge-conjugation compatibility from item 7. Replace "no component on the model kernel" by $\Pi(\chi_c\zeta) = 0$, and distinguish the two action integrals in the tunnelling and defect estimates. The assertion that only the wall breaks C also needs the added connection hypothesis.
   - **Bimodule, line 185.** Specify "32 complex-dimensional bimodule with a real structure," equivalently 64 real dimensions before taking a real fixed subspace.
   - **Status, line 189.** Lemma 29 case (b) had a Terra referee report; the corrected case-(a) text had not had a second reading. Proposition 30.7 and Theorem 30.5 were new.

10. **AMENDMENT — Physics framing and named sources.** The abstract acknowledges that the disc-bundle vector poles remain open, and the body says that **4** is pseudoreal. Put both qualifications next to the headline claim: the disc-bundle result treats a **fermion** window, and no calculation identifies its tunnelling scale with M_W or M_Z. The title should describe a target electroweak limit or a fermion joint limit. The auxiliary $\hat{\Gamma}$-grading and the zero virtual $Sp(2)$-index do not establish the chiral Standard Model spectrum.

The reading of Witten's October 1983 preprint, pp. 7–8 and 43–44, is consistent at the stated level. Witten

discusses vectorlike quantum numbers under the specified geometric gauge assumptions and states equivariant cobordism invariance of the character-valued index. Here the extending object is the **sum** of the four line summands; the individual $\mathcal{O}(k)$ do not all extend. The signed indices are $(4, 0, 0, -4)$. This index permits the spectral pairing and does not prove the window.

Maridakis, arXiv:2110.11654v2, §2, Assumption 3 requires a direction-independent leading endomorphism. For spin 3/2, $(n\cdot J)^2$ depends on n as an endomorphism, even though its eigenvalues are direction-independent. The paper also works on a closed manifold, so the boundary layer requires separate analysis.

The Bryant–Salamon qualification is correct. Karigiannis–Lotay, arXiv:2002.06444v3, Eq. (3.36) and Theorem 3.6 gives a complete asymptotically conical G_2 metric on the same total bundle, with a radial horizontal scale and non-flat fibre metric outside Setting 29.0. Homothetic enlargement of the fixed shape has a flat pointed limit at a fixed interior point; a fixed boundary point gives a half-space limit.

11. **AMENDMENT — Citations and companion drafts.** The internal interval references support the exact spectrum and the qualified Taylor coefficient. The note should preserve the companion draft's explicit distinction between a leading Taylor quotient and a pole calculation. towerless_intervals.tex treats the relation between the KO labels and geometric limits as a proposal; the joint note's introduction presents that relation more definitively and should retain the proposal status.

Connes–van Suijlekom, §1 supports the operator-system discussion. Replace "generated algebras do not converge when compression maps do" at line 146 by "need not converge"; constant families already show that convergence can occur.

The Kirchberg citation is apt for the **untwisted** $R/3$ consistency check in complex dimension three. It does not by itself prove the twisted $\mathcal{O}(k)$ spectrum. Kirchberg's 1990 paper, introduction and Theorem 2, explicitly gives the odd-dimensional bound and identifies $\mathbb{CP}^3$ as a limiting case. The Parthasarathy and Bismut–Lebeau entries have matching bibliographic identities in the Annals record and Numdam record, respectively; their full original texts could not be checked directly in this review, so the precise source-level attribution remains unchecked here. The numerical normalization in lemma 29 must be justified in the argument.

Baptista, arXiv:2506.09126v2, §1 and Eq. (1.2), supports the distinction between Killing and non-Killing metric modes in its stated Kaluza–Klein setting. It does not calculate the vector sector on this disc bundle.

12. **AMENDMENT — Table I.** Every displayed $M = 32$ row matches lemma 29's Python table. The first **16** entry is the core value and the second is the wall value; on the **20**, the first is the wall value and the second is the core value. Adding those labels would make the comparison easier to assess.

The smallest computed **4** value is floor-dominated at the reported meshes. Its mesh dependence does not determine whether the continuum value vanishes. The stated 0.005 wall discrepancy is a Python mesh observation from the source, not an independently established error bound. The caption should attribute the limit values to the end models and their spectral comparison: the window theorem by itself gives the low-window count and gap, not all individual excited limits.

13. **AMENDMENT — Prior results and comparison.** Several mechanisms in the note have established precedents that an expert would expect to see distinguished from the new combination.

- Kaplan, hep-lat/9206013v1, Eqs. (0) and surrounding discussion, derives chiral defect localization. Shamir, hep-lat/9303005v1, §§1–3, treats boundary fermions and the finite fifth direction. These precede the interval localization mechanism.
- Csáki et al., hep-ph/0305237v1, §§2 and 6, derives gauge symmetry breaking by interval boundary conditions. Cacciapaglia–Csáki–Park, hep-ph/0510366v2, §5, obtains $m_t = 2M_W$ at tree level for a suitable rank-four representation. Neither source establishes the lemma-25 crossover.
- Maridakis, arXiv:1510.07004v1, Lemma 2.2, gives the concentrating-pair square identity. Kaad–van Suijlekom, arXiv:1710.03182v1, Proposition 18, gives a horizontal–vertical Dirac factorization with a curvature contraction. Their settings differ from the stated seven-dimensional boundary problem.
- Kottke, arXiv:0909.5661v2, §5, gives a families Callias index theorem under asymptotically conic hypotheses. It supplies index information under those hypotheses; the finite boundary-window estimate

is a separate question.

- Iso–Kitazawa–Suyama, arXiv:2111.06508v2, §§1 and 4–5, follows massive Kaluza–Klein gauge modes that become massless during symmetry restoration in flux compactifications. The mechanism and geometry differ from the present joint family.

Witten's closed-geometry index restrictions and the boundary construction have different hypotheses; no contradiction was found in the checked records. These targeted checks do not establish priority for the eight-state theorem.

Corrections in order of severity

1. Add global charge-conjugation compatibility to Proposition 30.7 and the note, or supply a different proof for the allowed case-(b) interpolations. Retain the unconditional $O(M^{-1/2})$ defect estimate until then.
2. Remove the claimed implication from the hierarchy alone to divergent fixed-energy counting; retain the separate counts proved for the named families.
3. Correct Proposition 29.10's signed fibre part to $-w(\rho)$, and add the projection estimate for the adjoint wall bound.
4. Restore the disc-bundle hypotheses in the note, restrict the cylinder data, and give the localized interval family its own length.
5. Qualify the electroweak, chirality, Taylor-mass and finite-versus-limit compression claims in the title, abstract and theorem statements.

Part R18 (assessment of the Astra audit)

Assessment of the audit of commit b2ddf98

The user supplied on 4 November 2026 an outside audit of the repository at b2ddf98, titled "conneskk: audit and a route from the massless electroweak limit to eleven dimensions", kept verbatim in 2026-11-03_astra_audit_d11.md. The file does not state its authorship; the 11D line, merged on 4 November, attributes it to Astra 6, and we follow that. This note records what was checked and what was done. Provenance: Claude Sonnet 5.5 recomputed all items, read-only, in Python and with pdftotext, and Claude Opus 5.5 rederived by hand the items marked below, against the source lemmas. The user asked for attention to detail with the reports of smaller models, so the column "by hand" says which verdicts do not rest on such a report alone.

Verdicts

Audit section	Claim	Verdict	By hand
1	Withdrawal of the alleged normalization error at boundary_condition_note.tex:53	Correct: both components have the factor	Sonnet only
2.1	D^2 = 2 lambda^2 on the 64-dimensional product, P_- D P_- = 0,		[P_-, D]
2.2	The massless limit must specify a path; theta -> 0 at fixed l opens no tower	Correct, and adopted as STRATEGY T14	Yes
2.3	ind D_{O(k)} = (k^3 - k)/6, values 4, 0, 0, -4, total zero	Correct	Yes; Witten 1983, pp. 43-44, gives the reason
2.4	Supplement omits lemma 17	Correct; lemmas 01, 02, 04 are also omitted, and 17 is added	Yes

Audit section	Claim	Verdict	By hand
2.4	docs/ last refreshed at 4ae5b43; T13 steers only to D = 9; docs/index.html has no seven-dimensional entry	Correct	Sonnet only
2.4	PDF overflow, 81 of 478 and 13 of 263 pages	Depends on an unstated tolerance: Sonnet found 113 and 15 at tolerance 0, 83 and 15 at 2 pt, 12 mobile pages above 5 pt, with maximum overhang 8.7 pt, through pdftotext -bbox	Sonnet only
3	Finite window and opening tower on the interval, equations (1) to (4)	Correct; the statement needs Delta_eps defined, and the references are Theorems 22.1, 22.4. Now Theorem 30.1	Yes
3.1	Vector poles and M_W^2 = 2 m^2, M_Z^2 = (16/5) m^2; coefficients 3 and 24/5	Correct, the latter for the fully massive family of lemma 18, Proposition 14. Moreover M^2 = ((3 - d_eff)/2)	
4	Localized family, equations (6) to (8), and the table	Correct. Now Proposition 30.3	Yes; table rows by Sonnet
5.1	Equation (9), Weyl exponent on the cylinder	Correct. Now Corollary 30.4	Yes
5.2	Dilation, conditional scaling of X	Correct; the condition is removed in Theorem 30.5 for the class of lemma 29, pending its referee	Yes
6	Bound (13) with constant 2, and the telescoping	Correct, once “defect” means	
6, (14)	Target: eta_t <= C (lambda/delta)^2 + errors	Stated as a target there. Proposition 30.7 proves the form exp(-2 G) for the upper bound exp(-G) on lambda; a bound in terms of lambda itself would need a lower bound on lambda, which is open	Yes
6.1	Table of eta_t at L = 2, 3, 4, 5	Rows reproduced by Sonnet (Python, same solver, so a reproduction with a new observable)	Sonnet only

Audit section	Claim	Verdict	By hand
7.1	Toy potential and its minimum	Correct as an elementary statement; underived, as the audit says	Yes
Appendix A	M^{111} data: c_1 = 3h + 2u, dim H^0(O(3,2)) = 30; six arXiv identifiers	Correct on the items checked	Sonnet only

No error in a displayed equation of the audit was found.

What was done

- Lemma 30 takes the propositions that are proved, with proofs redone, and adds Theorem 30.5 and Proposition 30.7.
- Item 2 of the audit's list (the shape-unit estimate for X, starting from the line-preserving collar) is lemma 29, which was already in progress when the audit arrived, and goes further: case (a) at the Fubini-Study radius as well.
- STRATEGY gains T14, the joint limit, parallel to T13, as the audit asks.
- paper/build_supplement.sh gains lemma 17, and lemmas 28 to 30 once their referees are applied.

What was not adopted

The note proposed in section 8 is being written at the request of the user, as paper/joint_limit_note.tex. The M^{111} comparison stays a comparison, and no W, Z -> 0 family links it to the finite geometry. The layout repair of the supplement is recorded and has not been done.